IBNR reserving

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Introduction

- In this chapter, we will discuss the methods for calculating the amount of provisions for outstanding claims, and more particularly the methods for quantifying the associated error.

Introduction

- Technical provisions are financial reserves that allow full settlement of commitments to policyholders and policy beneficiaries. They are linked to the same technique of insurance and imposed by regulations.

Introduction

- From formal terms, at time t, the insurance company is required to make a provision for claims incurred before the date it will be under obligation to indemnify.

Introduction

- It must estimate the cost of claims incurred, and subtract the amounts already paid. So this is basically a forecasting problem.

The problem of reserving

- Among the methods recognized by the authorities of control, the most classic is based on payment rates. We proceed per year of occurrence of a disaster, and it is assumed certain regularity in the rate of payment.

The problem of reserving

- Conventionally, it will be noted;
  - i (in line) the year of the sinister occurrence;
  - j (in column) development year;
  - $i + j$ (diagonally) the calendar year of payment (for increments);

The problem of reserving

- $Y_{i,j}$ payments increments for development year j, for incidents occurring in year *i* (see Table 1);
- $C_{i,j}$ cumulative payments, in the sense that $\boldsymbol{C_{i,j} = Y_{i,0} + Y_{i,1} + \cdots + Y_{i,j}}$ for accident year *i* (see Table 2);
- P_i the earned premium for the year *i* (see Table 3);

The problem of reserving

- $N_{i,j}$ the cumulative number of claims for accident year i saw at the end of j years (see Table 4);
- $\Gamma_{i,j}$ cumulative charge file by file, for the development year j, for incidents occurring in year i (see Table 7);

The problem of reserving

	0	1	2	3	4	5
0	3209	1163	39	17	7	21
1	3367	1292	37	24	10	
2	3871	1474	53	22		
3	4239	1678	103			
4	4929	1865				
5	5217					

Table 1 - Triangle of the payments increments, $Y = (Y_{i,j})$

The problem of reserving

	0	1	2	3	4	5
0	3209	4372	4411	4428	4435	4456
1	3367	4659	4696	4720	4730	
2	3871	5345	5398	5420		
3	4239	5917	6020			
4	4929	6794				
5	5217					

Table 2 - Triangle of cumulative payments, $C = (C_{i,j})$

The problem of reserving

Year i	0	1	2	3	4	5
P_i	4591	4672	4863	5175	5673	6431

Table 3 - Earned premiums vector, $P = (P_i)$

The problem of reserving

	0	1	2	3	4	5
0	1043.4	1045.5	1047.7	1047.7	1047.7	1047.7
1	1043.0	1027.1	1028.7	1028.9	1028.7	
2	965.1	967.9	967.8	970.1		
3	977.0	984.7	986.8			
4	1099.0	1118.5				
5	1076.3					

Table 4 - Triangle of numbers of claims, cumulated, in thousands, $N = (N_{i,j})$

The problem of reserving

- Formally, all this data is stored in matrices (or a vector for the premium), with missing values noted `NA` for future values. They will be designated respectively `PAID`, `PREMIUM`, `NUMBER` and `INCURRED`.

The problem of reserving

```
> PAID <-
matrix(c(3209,4372,4411,4428,4435,4456,3367,4659,46
96,4720,4730,NA,3871,5345,5398,5420,NA,NA,4239,5917
,6020,NA,NA,NA,4929,6794,NA,NA,NA,NA,5217,NA,NA,NA,
NA,NA), nrow = 6, ncol = 6, byrow = T)

> PAID
```

The problem of reserving

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]
[1,]	3209	4372	4411	4428	4435	4456
[2,]	3367	4659	4696	4720	4730	NA
[3,]	3871	5345	5398	5420	NA	NA
[4,]	4239	5917	6020	NA	NA	NA
[5,]	4929	6794	NA	NA	NA	NA
[6,]	5217	NA	NA	NA	NA	NA

The problem of reserving

- The triangle increments are easily deduced from the cumulative triangle:

```
> nc <- ncol(PAID)
> nl <- nrow(PAID)
> INC <- PAID
> INC[,2:nc] <- PAID[,2:nc] - PAID[,1:(nc-1)]
> INC
```

The problem of reserving

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]
[1,]	3209	1163	39	17	7	21
[2,]	3367	1292	37	24	10	NA
[3,]	3871	1474	53	22	NA	NA
[4,]	4239	1678	103	NA	NA	NA
[5,]	4929	1865	NA	NA	NA	NA
[6,]	5217	NA	NA	NA	NA	NA

The problem of reserving

- In its simplest version, the purpose of provisioning methods is to complete the lower part of the payments of triangles.

The problem of reserving

- Provisioning is a prediction problem, conditioned by information available at the time *n*. Note $\mathcal{H}_n$ information available at the time *n*, that to say

$$\boldsymbol{\mathcal{H}_n = \left\{\left(X_{i,j}\right), i + j \leq n\right\} = \left\{\left(C_{i,j}\right), i + j \leq n\right\}}$$

The problem of reserving

- It seeks to study, by accident year, the conditional distribution of $C_{i,\infty}$ (the ultimate burden for a given year of occurrence), knowing $\mathcal{H}_n$, or, assuming the sinister closed after n years the distribution of $C_{i,n}$ knowing $\mathcal{H}_n$. If we focus on a particular accident year, we may note

$$\boldsymbol{\mathcal{F}_{i,n-i} = \left\{\left(X_{i,j}\right), j = 0, \dots, n-i\right\} = \left\{\left(C_{i,j}\right), j = 0, \dots, n-i\right\}}$$

The problem of reserving

- This notation allows taking into account the available information changes from one line to another. We seek thereafter to predict the amount of claims paid for the year i

$$\widehat{\boldsymbol{C}}_{\boldsymbol{i,n}}^{(\boldsymbol{n-i})} = \mathbb{E}\left[\boldsymbol{C}_{\boldsymbol{i,n}} | \boldsymbol{\mathcal{F}}_{\boldsymbol{i,n-i}}\right]$$

The problem of reserving

- And the difference between that amount and the amount already paid will be the provision for incurred but not reported loss

$$\widehat{R}_i = \widehat{C}_{i,n}^{(n-i)} - C_{i,n-i}$$

The problem of reserving

- Then we try to quantify uncertainty in this prediction. As we will see the usual methods were designed to calculate

$$\mathbb{V}\left[\boldsymbol{C_{i,n}}|\boldsymbol{\mathcal{F}_{i,n-i}}\right] \boldsymbol{or}\ \mathbb{V}\left[\widehat{\boldsymbol{C}}_{\boldsymbol{i,n}}^{(\boldsymbol{n-i})}\right]$$

The problem of reserving

- And more particularly the change in the estimate of the ultimate load

$$\Delta_i^n = \widehat{C}_{i,n}^{(n-i+1)} - \widehat{C}_{i,n}^{(n-i)} = CDR_i(n)$$

The problem of reserving

- Sometimes noted CDR (claims development result). If this difference is positive, it is called mali, and if negative, it is called boni. It can be shown that

$$\mathbb{E}\left[\Delta_{\boldsymbol{i}}^{\boldsymbol{n}} | \boldsymbol{\mathcal{F}}_{\boldsymbol{i},\boldsymbol{n}-\boldsymbol{i}}\right] = \mathbf{0}$$

The problem of reserving

- In other words, we cannot hope to do neither boni nor mali, on average. Regulatory constraints imposed by Solvency II require computing

$$\mathbb{V}\left[\Delta_{\boldsymbol{i}}^{\boldsymbol{n}} | \boldsymbol{\mathcal{F}}_{\boldsymbol{i},\boldsymbol{n}-\boldsymbol{i}}\right]$$

Payments cadences and *Chain-Ladder* method

- The use of payment rates to estimate the future charge existed since 1930. It is assumed that there is a form of recursive relationship

$$\boldsymbol{C_{i,j+1} = \lambda_j \times C_{i,j}}$$

- For any

$$\boldsymbol{i, j = 1, \dots, n}$$

Payments cadences and *Chain-Ladder* method

- A natural estimator for λ_j, based on past experience is then the average ratio based on $n - j$ years observed

$$\hat{\lambda}_j = \frac{\sum_{i=1}^{n-j} C_{i,j+1}}{\sum_{i=1}^{n-j} C_{i,j}}$$

- For any

$$j = 1, \dots, n-1$$

Payments cadences and *Chain-Ladder* method

- So that it is then possible to predict the burden for unobserved in the triangle part

$$\widehat{C}_{i,j} = \left[\hat{\lambda}_{n+1-i} \dots \hat{\lambda}_{j-1}\right] \times C_{i,n+1-i}$$

Payments cadences and *Chain-Ladder* method

```
> k <- 1
> weighted.mean(x = PAID[,k+1]/PAID[,k], na.rm
= TRUE)
[1] 1.380933
>  sum(PAID[1:(nl-k), k+1])/sum(PAID[1:(nl-k),
k])

[1] 1.380933
```

Payments cadences and *Chain-Ladder* method

```
> LAMBDA <- rep(NA, nc-1)
> LAMBDA
> for(k in 1:(nc-1)) {
LAMBDA[k] = (sum(PAID[1:(nl-k), k+1])/sum(PAID[1:(nl-k), k]))}
> LAMBDA
[1] 1.380933 1.011433 1.004343 1.001858 1.004735
```

Payments cadences and *Chain-Ladder* method

- Note that instead of calculating development factors, we can also calculate growth rates, cumulative or not. In other words, instead of writing

$$C_{i,j+1} = \lambda_j \times C_{i,j}$$

- For any

$$i, j = 1, \dots, n$$

Payments cadences and *Chain-Ladder* method

- Note that

```
> (GAMMA <- rev(cumprod(rev(1/LAMBDA))))
[1] 0.7081910 0.9779643 0.9891449 0.9934411
0.9952873
> (PHI <- c(GAMMA[1], diff(GAMMA)))
[1] 0.708191033 0.269773306 0.011180591
0.004296183 0.001846141
```

Payments cadences and *Chain-Ladder* method

	0	1	2	3	4	n
λ_j	1.380933	1.011433	1.004343	1.001858	1.004735	1.0000
γ_j	70.819%	97.796%	98.914%	99.344%	99.529%	100%
ϕ_j	70.819%	26.977%	1.118%	0.430%	0.185%	0.000%

Table 5 - Factors of development, $\hat{\lambda} = (\hat{\lambda}_i)$, expressed as a rate of payments compared to the ultimate load, on aggregate (i.e $\hat{\gamma}$), then in increments (i.e $\hat{\varphi}$)

Payments cadences and *Chain-Ladder* method

- This coefficient allows visualizing the payment rate (percentage of the ultimate load). We can visualize this ratio in Figure 1.

```
> barplot(PHI, names = 1:5)
```

Payments cadences and *Chain-Ladder* method

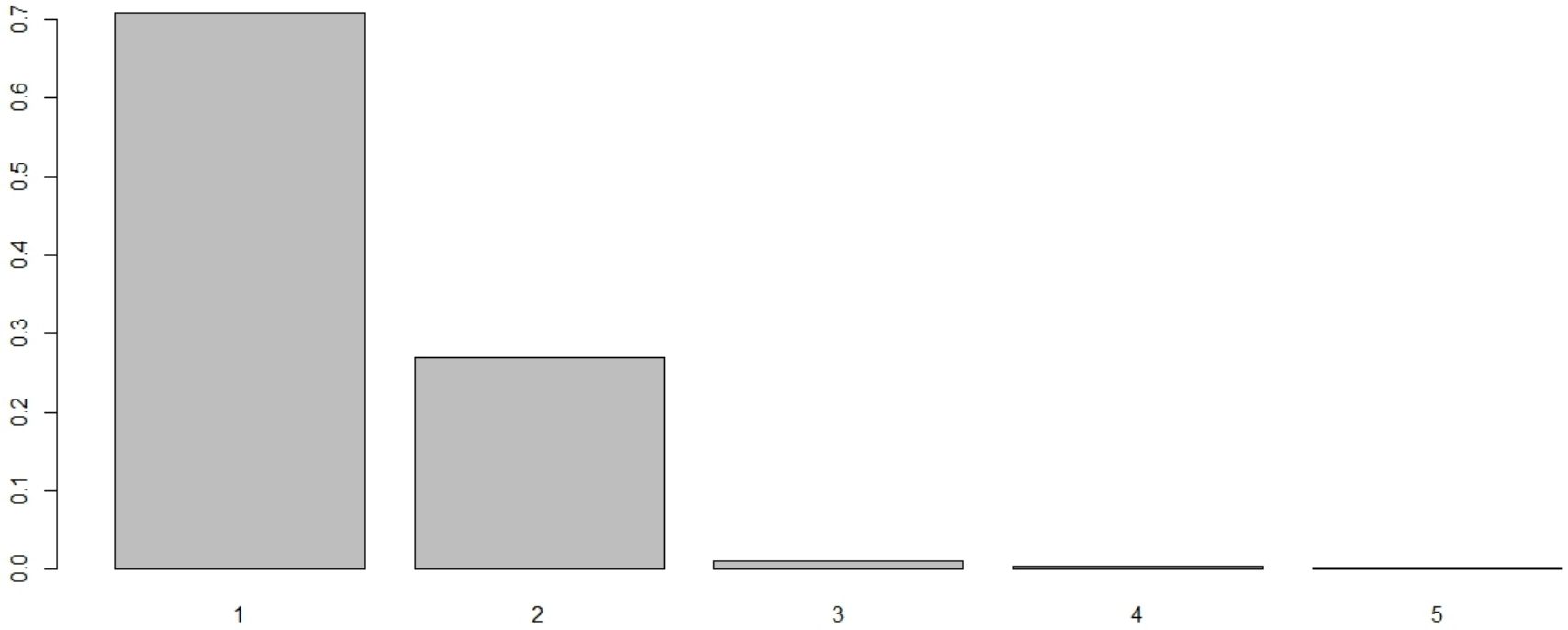


Figure 1 – Claims Paying rate, depending on the ultimate load, the first year (calendar year of occurrence of the accident) to the fifth year

Payments cadences and *Chain-Ladder* method

- Note that it is possible to see the Chain Ladder estimator $\hat{\lambda}_j$ as a weighted average of the individual transition factors, i.e.

$$\hat{\lambda}_j = \sum_{i=1}^{n-j} \omega_{i,j} \times \lambda_{i,j}$$

Payments cadences and *Chain-Ladder* method

- Where

$$\omega_{i,j} = \frac{C_{i,j}}{\sum_{i=1}^{n-j} C_{i,j}}$$

- And

$$\lambda_{i,j} = \frac{C_{i,j+1}}{C_{i,j}}$$

Payments cadences and *Chain-Ladder* method

- Also, we can get these coefficients using weighted linear regressions without constant, by regressing the $C_{.,j+1}$ on $C_{.,j}$. Thus, for the first value

```
> lm(PAID[,k+1] ~ 0 + PAID[,k], weights =
1/PAID[,k])$coefficients
PAID[, k]
1.004735
```

Payments cadences and *Chain-Ladder* method

- Once estimated the development factor, nothing is easier than to complete the triangle, always iterating, column after column:

```
> TRIANGLE <- PAID
> for (i in 1:(nc-1)) {
TRIANGLE[(nl-i+1):(nl), i+1] =
LAMBDA[i]*TRIANGLE[(nl-i+1):(nl), i]}
> TRIANGLE
```

The problem of reserving

	0	1	2	3	4	5
0	3209	4372.000	4411.000	4428.000	4435.000	4456.000
1	3367	4659.000	4696.000	4720.000	4730.000	4752.397
2	3871	5345.000	5398.000	5420.000	5430.072	5455.784
3	4239	5917.000	6020.000	6046.147	6057.383	6086.065
4	4929	6794.000	6871.672	6901.518	6914.344	6947.084
5	5217	7204.327	7286.691	7318.339	7331.939	7366.656

Table 6 - Triangle of cumulative payments, $C = (C_{i,j})_{i+j\leq n}$ with their future projection $\widehat{C} = (\widehat{C}_{i,j})_{i+j>n}$

Payments cadences and *Chain-Ladder* method

- The amount of provisions is the difference between what we predict to pay for each accident year and what we have already paid.

```
> facteur = 1
> chargeultime <- TRIANGLE[,nc]*facteur
> chargeultime
[1] 4456 4752 5455 6086 6947 7366
```

Payments cadences and *Chain-Ladder* method

```
> paiements <- diag(TRIANGLE[,nc:1])
[1] 4456 4730 5420 6020 6794 5217
> (RESERVES <- chargeultime - paiements)
[1]    0.00000   22.39684   35.78388   66.06466
153.08358 2149.65640
```

Payments cadences and *Chain-Ladder* method

- It is noted here sum (RESERVED) worth (2426.985), which is the total amount of reserves that should be allocated. A faster algorithm is to directly use the formula based on the product transition ratios.

Payments cadences and *Chain-Ladder* method

- Was then

```
> DIAG <- diag(TRIANGLE[,nc:1])
> PRODUIT <- c(1, rev(LAMBDA))
> sum((cumprod(PRODUIT)-1)*DIAG)
[1] 2426.985
```

Payments cadences and *Chain-Ladder* method

- For the rest, we can develop `Chainladder()` function that returns the ultimate burden per line and reserves

```
> Chainladder <- function(TR, f=1) {
> nc <- ncol(TR); nl <- nrow(TR)
> L <- rep(NA, nc-1)
> for(k in 1:(nc-1)) {
L[k] = lm(TR[, k+1] ~ 0 + TR[,k], weights =
1/TR[,k])$coefficients}
```

Payments cadences and *Chain-Ladder* method

```
> TRc <- TR
> for(i in 1:(nc-1)) {
TRc[(nl-i+1):(nl), i+1] = L[i] * TRc[(nl-
i+1):(nl), i]}
> C <- TRc[,nc]*f
> R <- (cumprod(c(1, rev(L)))-
1)*diag(TR[,nc:1])
> return(list(charge = C, reserves = R, restot
= sum(R))) }
```

Payments cadences and *Chain-Ladder* method

```
> Chainladder(PAID)
$charge
[1]    4456.000    4752.397    5455.784    6086.065
6947.084 7366.656
$reserves
[1]     0.00000    22.39684    35.78388    66.06466
153.08358 2149.65640
$restot
[1] 2426.985
```

From Mack to Merz & Wüthrich

- The said *Chain Ladder* method, which we have just seen, is a deterministic method known in the sense that we do not build probabilistic model to measure the uncertainty associated with the prediction of the amount of reserves.

From Mack to Merz & Wüthrich

- Different models have been proposed from the 90s, from the model of Mack up the approach proposed by Merz & Wüthrich which will introduce the concept of uncertainty to one year.

From Mack to Merz & Wüthrich

- We got through the chain ladder method, an estimator of the amount of provision $\hat{R}$. Conventionally, to quantify the error associated with an estimator, we calculate the mean squared error - mse - associated.

$$\mathbb{E}\left(\left[\widehat{\boldsymbol{R}} - \boldsymbol{R}\right]^{\mathbf{2}}\right)$$

From Mack to Merz & Wüthrich

- Formally, *R* is here as a random variable, we do not talk of mse but prediction mse, noted msep (it is not predicted on past data, but the data will be used to calibrate a model that will then be used to make prediction for future years).

$$\boldsymbol{msep}(\widehat{\boldsymbol{R}}) = \mathbb{E}\left(\left[\widehat{\boldsymbol{R}} - \boldsymbol{R}\right]^{\mathbf{2}}\right)$$

From Mack to Merz & Wüthrich

- This term can be divided into two parts (by a first order approximation), in that

$$\mathbb{E}\left(\left[\widehat{\boldsymbol{R}}-\boldsymbol{R}\right]^{\mathbf{2}}\right) \simeq \underbrace{\mathbb{E}\left(\left[\widehat{\boldsymbol{R}}-\mathbb{E}(\boldsymbol{R})\right]^{\mathbf{2}}\right)}_{\boldsymbol{mse}\left(\widehat{\boldsymbol{R}}\right)} + \underbrace{\mathbb{E}\left(\left[\boldsymbol{R}-\mathbb{E}(\boldsymbol{R})\right]^{\mathbf{2}}\right)}_{\mathbb{V}(\boldsymbol{R})}$$

From Mack to Merz & Wüthrich

- Hence the left term is the estimation error, given the fact that we had to estimate the amount of provisions from the top of the triangle, and the term on the right is the classic error model (any model with a residual portion orthogonal to the observations and therefore unpredictable).

From Mack to Merz & Wüthrich

- In fact, strictly speaking, it rather seeks to calculate a conditional msep to the information available at the end of *n* years,

$$\boldsymbol{msep_n}(\widehat{\boldsymbol{R}}) = \mathbb{E}\left(\left[\widehat{\boldsymbol{R}} - \boldsymbol{R}\right]^{\boldsymbol{2}} | \boldsymbol{\mathcal{H}_n}\right)$$

From Mack to Merz & Wüthrich

- Mack proposed a probabilistic framework to justify the use of the *Chain Ladder* method. For this, it is assumed that

$$\left(C_{i,j}\right)_{j\geq 0}$$

From Mack to Merz & Wüthrich

- Is a Markov process, and there is

$$\boldsymbol{\lambda} = \boldsymbol{\lambda_j}$$

- And

$$\boldsymbol{\sigma} = \left(\boldsymbol{\sigma_j^2}\right)$$

From Mack to Merz & Wüthrich

- Such as

$$\begin{cases} \mathbb{E}\left(\boldsymbol{C_{i,j+1}}|\boldsymbol{\mathcal{H}_{i+j}}\right) = \mathbb{E}\left(\boldsymbol{C_{i,j+1}}|\boldsymbol{C_{i,j}}\right) = \boldsymbol{\lambda_j} \times \boldsymbol{C_{i,j}} \\ \boldsymbol{Var}\left(\boldsymbol{C_{i,j+1}}|\boldsymbol{\mathcal{H}_{i+j}}\right) = \mathbb{V}\left(\boldsymbol{C_{i,j+1}}|\boldsymbol{C_{i,j}}\right) = \boldsymbol{\sigma_j^2} \times \boldsymbol{C_{i,j}} \end{cases}$$

From Mack to Merz & Wüthrich

- We note under these assumptions

$$\mathbb{E}\left(\boldsymbol{C_{i,j+k}}|\boldsymbol{\mathcal{H}_{i+j}}\right) = \mathbb{E}\left(\boldsymbol{C_{i,j+k}}|\boldsymbol{C_{i,j}}\right) = \boldsymbol{\lambda_j} \times \boldsymbol{\lambda_{j+1}} \times \cdots \times \boldsymbol{\lambda_{j+k-1}} \times \boldsymbol{C_{i,j}}$$

From Mack to Merz & Wüthrich

- Mack adds another hypothesis of independence between accident years, ie

$$\left(C_{i,j}\right)_{j=1,\ldots,n}$$

- And

$$\left(C_{i',j}\right)_{j=1,\ldots,n}$$

- Are independent for any i ≠i'.

From Mack to Merz & Wüthrich

- A rewrite of the model is then to assume that

$$\boldsymbol{C_{i,j+1} = \lambda_j \times C_{i,j} + \sigma_j \times \sqrt{C_{i,j}} \times \varepsilon_{i,j}}$$

- Where residuals ($\varepsilon_{i,j}$) are independent and identically distributed, centered and unit variance.

From Mack to Merz & Wüthrich

- From this scripture, it may seem legitimate to use the weighted least squares methods for estimating these coefficients, noting that the weights must be inversely proportional to the variance, in other words the $C_{i,j}$, i.e. to j given, we seek to solve

$$\boldsymbol{min}\left\{\sum_{i=1}^{n-j}\frac{1}{C_{i,j}}\left(C_{i,j+1}-\lambda_j\times C_{i,j}\right)^2\right\}$$

From Mack to Merz & Wüthrich

- To test these first two hypotheses, we begin by representing the $C_{.,j+1}$ depending on $C_{.,j}$ given to j. If the first hypothesis is confirmed, the points must be aligned in a straight line through the origin.

From Mack to Merz & Wüthrich

```
> par(mfrow = c(1,2))
> j = 1
> plot(PAID[,j], PAID[,j+1], pch = 19, cex =
1.5)
> abline(lm(PAID[,j+1] ~ 0 + PAID[,j], weights
= 1 / PAID[,j]))
> j = 2
```

From Mack to Merz & Wüthrich

```
> plot(PAID[,j], PAID[,j+1], pch = 19, cex =
1.5)
abline(lm(PAID[,j+1] ~ 0 + PAID[,j], weights =
1 / PAID[,j]))
```

- Figure 2 used to display the payments steady rate hypothesis in time. The regression is weighted with the same weight as those used to estimate the transition regression coefficients.

From Mack to Merz & Wüthrich

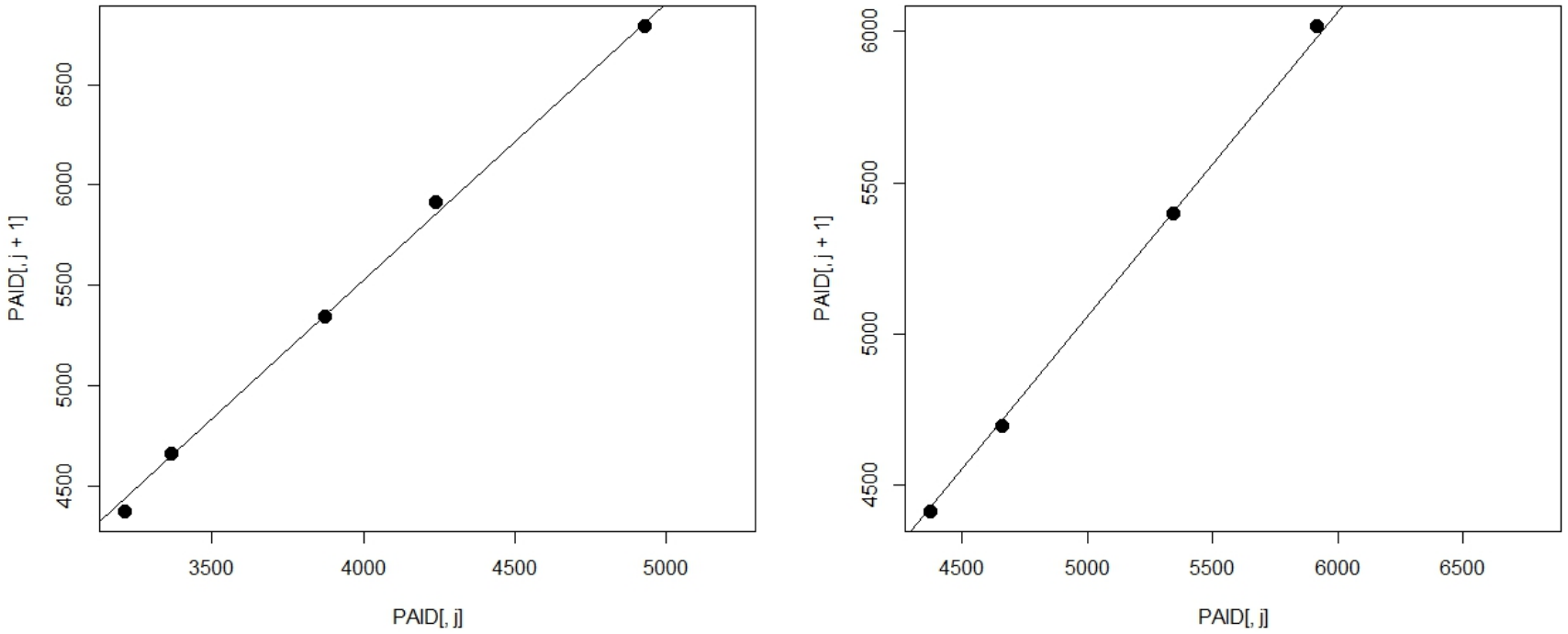


Figure 2 – Cloud for $C_{.j+1}$ depending on $C_{.,j}$ for j = 1 to the left, and j = 2 to the right. The regression line through the origin and weights are inversely proportional to the amount of payments

From Mack to Merz & Wüthrich

- For the second, the standardized residuals can be studied (Mack speaks of weighted residuals).

$$\boldsymbol{\varepsilon}_{i,j} = \frac{\boldsymbol{C_{i,j+1} - \hat{\lambda}_j \times C_{i,j}}}{\sqrt{\boldsymbol{C_{i,j}}}}$$

From Mack to Merz & Wüthrich

- The use of standardized residuals also gives us a simple idea to estimate the volatility parameter.

$$\widehat{\sigma}_j^2 = \frac{1}{n-j-1} \sum_{i=0}^{n-j-1} \left(\frac{C_{i,j+1} - \widehat{\lambda}_j \times C_{i,j}}{\sqrt{C_{i,j}}} \right)^2$$

From Mack to Merz & Wüthrich

- This can also be written

$$\widehat{\sigma}_j^2 = \frac{1}{n-j-1} \sum_{i=0}^{n-j-1} \left(\frac{C_{i,j+1}}{C_{i,j}} - \widehat{\lambda}_j \right)^2 \times C_{i,j}$$

- This is similar to the writing of the transition factor λ_j as a weighted average of the observed transitions factors.

From Mack to Merz & Wüthrich

```
> lambda <- PAID[,2:nc] / PAID[,1:(nc-1)]
> SIGMA <- rep(NA, nc-1)
> for(i in 1:(nc-1)){
> D <- PAID[,i] * (lambda[,i] -
t(rep(LAMBDA[i], nc)))^2
> SIGMA[i] <- 1 / (nc - i - 1) * sum(D[,1:(nc-
i)])}
```

From Mack to Merz & Wüthrich

```
> SIGMA[nc-1] <- min(SIGMA[(nc-3):(nc-2)])
> (SIGMA = sqrt(SIGMA))
[1] 0.72485777 0.32036422 0.04587297 0.02570564
0.02570564
```

- This method estimates the different parameters involved in the *Mack* model.

From Mack to Merz & Wüthrich

- Until now, we assume that the first line of our triangle is closed: there is no open claim, and therefore the provision amount for the year of occurrence is zero. This line is the basis for all subsequent developments.

From Mack to Merz & Wüthrich

- This assumption may be too high for long tail sector. Mack laid the foundation of the first models still used to overcome this assumption. Assume that there exists a $\lambda_\infty > 1$ as

$$\boldsymbol{C_{i,\infty} = C_{i,n} \times \lambda_\infty}$$

From Mack to Merz & Wüthrich

- A method (which has often been used) based on the idea that we could project the λ_i by an exponential extrapolation (or linear extrapolation of log (λ_k - *1*)), then we set

$$\boldsymbol{\lambda_\infty = \prod_{k \geq n} \hat{\lambda}_k}$$

- But better to be careful, especially if there are aberrant values.

From Mack to Merz & Wüthrich

```
> logL <- log(LAMBDA - 1)
> tps <- 1:(nc-1)
> modele <- lm(logL ~ tps)
> plot(tps, logL, xlim = c(1,20), ylim = c(-
30,0))
> abline(modele)
```

From Mack to Merz & Wüthrich

```
> tpsP <- seq(6,1000)
> logP <- predict(modele, newdata =
data.frame(tps = tpsP))
> points(tpsP, logP, pch = 0)
> (facteur <- prod(exp(logP) + 1))
[1] 1.000707
```

From Mack to Merz & Wüthrich

- In other words, this method expects to add 0.07% load compared to the prediction made by conventional methods, assuming close the first year. Numerically, this would amount to the provision

```
> chargeultime <- TRIANGLE[,nc] * facteur
> paiements <- diag(TRIANGLE[,nc:1])
```

From Mack to Merz & Wüthrich

```
> (RESERVES <- chargeultime - paiements)
[1]    3.148948   25.755248   39.639346
70.365538  157.992918 2154.862234
> sum(RESERVES)
[1] 2451.764
```

- Figure 3 is used to display the adjusted linear model on log transitional factors.

From Mack to Merz & Wüthrich

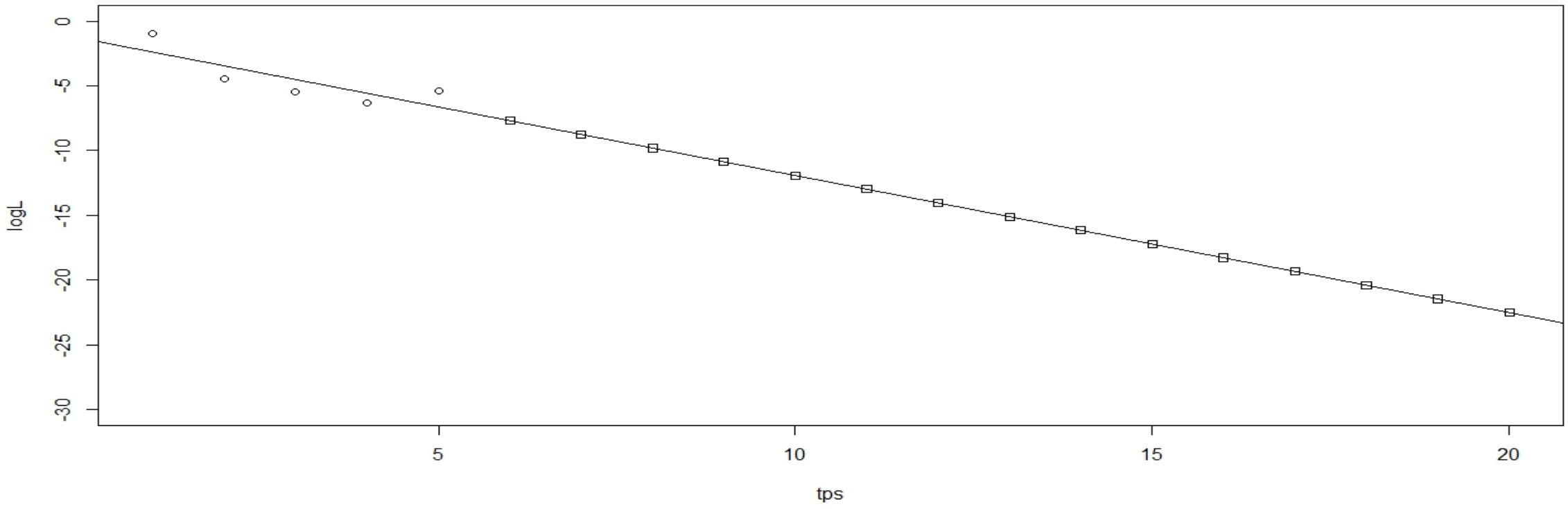


Figure 3 – Evolution of *log* (λ_j - *1)* and a linear prediction model

From Mack to Merz & Wüthrich

- From all these estimators, we can estimate the amount of the provision msep occurrence year, $\widehat{R}_i$, but also aggregated all years of combined occurrence. Numerically, we can use the function `MackChainLadder` of *library(ChainLadder)*.

From Mack to Merz & Wüthrich

```
> library(ChainLadder)
> MackChainLadder(PAID)
MackChainLadder(Triangle = PAID)
```

From Mack to Merz & Wüthrich

	Latest	Dev.To.Date	Ultimate	IBNR	Mack.S.E	CV(IBNR)
1	4,456	1.000	4,456	0.0	0.000	NaN
2	4,73	0.995	4,752	22.4	0.639	0.0285
3	5,42	0.993	5,456	35.8	2.503	0.0699
4	6,02	0.989	6,086	66.1	5.046	0.0764
5	6,794	0.978	6,947	153.1	31.332	0.2047
6	5,217	0.708	7,367	2,149.7	68.449	0.0318

From Mack to Merz & Wüthrich

	Totals
Latest:	32,637.00
Dev:	0.93
Ultimate:	35,063.99
IBNR:	2,426.99
Mack S.E.:	79.30
CV(IBNR):	0.03

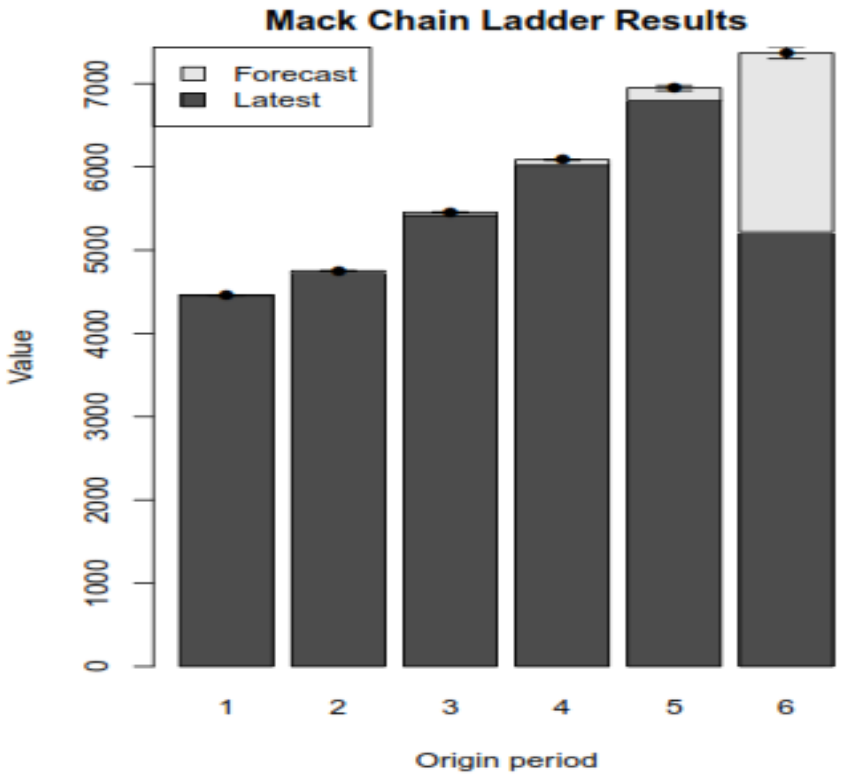
From Mack to Merz & Wüthrich

- We find the estimated total provisions $\widehat{\boldsymbol{R}}$, *IBNR*, worth 2,426.99, and msep($\widehat{\boldsymbol{R}}$) corresponding to that *Mack SE* here is 79.30. The occurrence year information's are shown in the first part of the table.

From Mack to Merz & Wüthrich

- We also obtained several graphics using the plot() function, corresponding to Figures 4, 5 and 6.

From Mack to Merz & Wüthrich



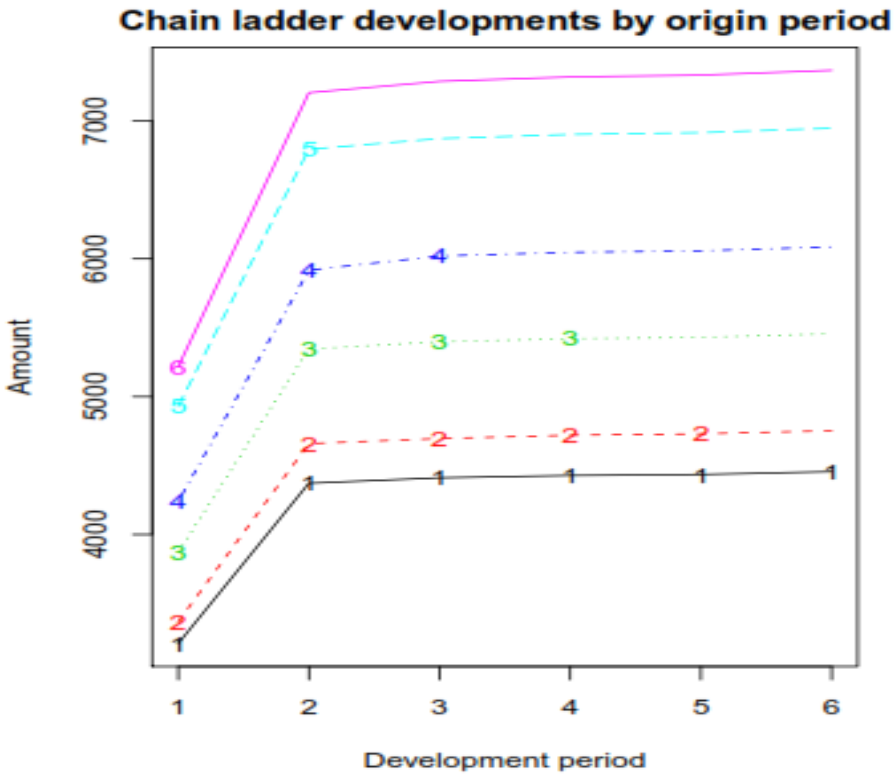


Figure 4 – Comparison between the estimated final charge and the sum already paid, left, and payment rates predicted by the chain ladder method

From Mack to Merz & Wüthrich

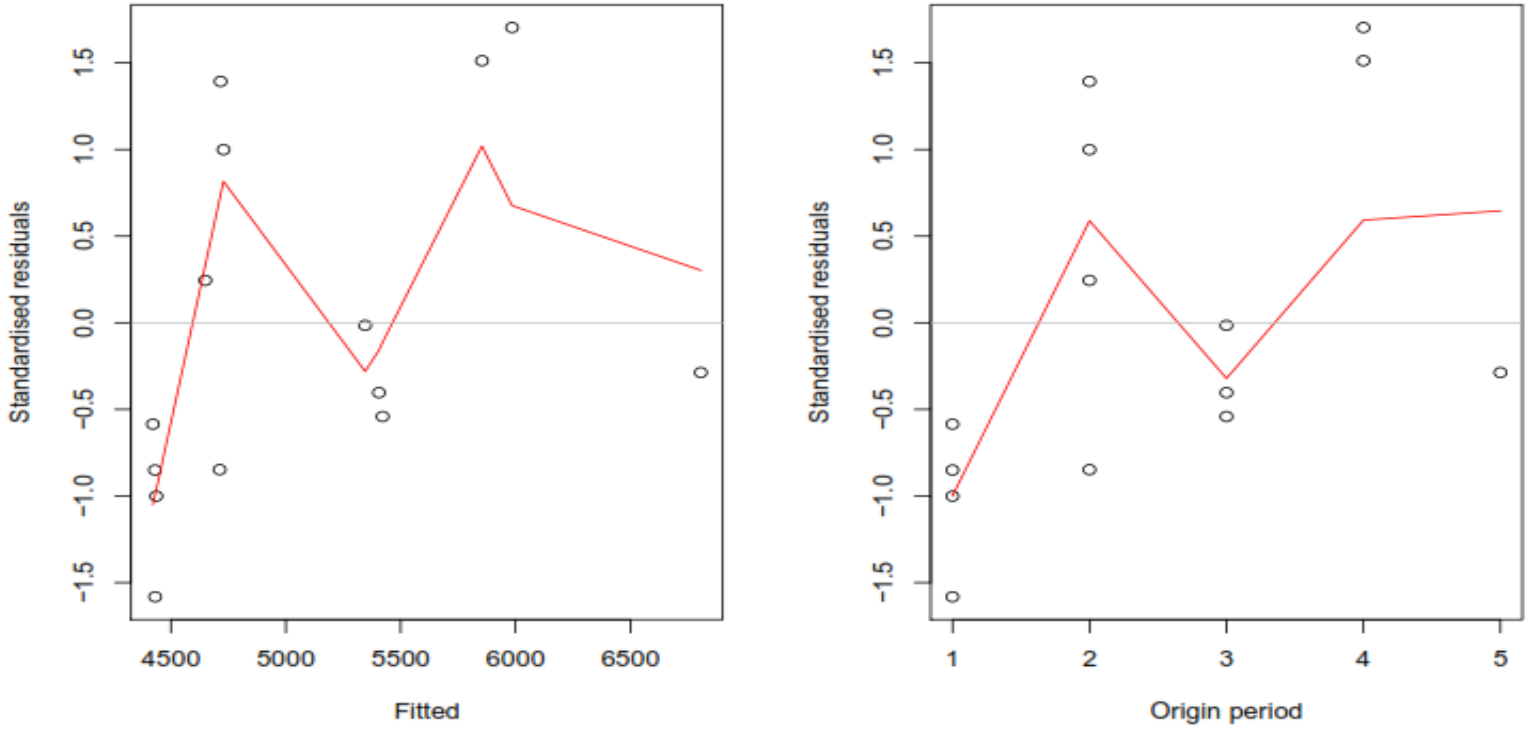


Figure 5 – Evolution of the standardized residuals according to $\widehat{C}_{i,j}$ and *i*

From Mack to Merz & Wüthrich

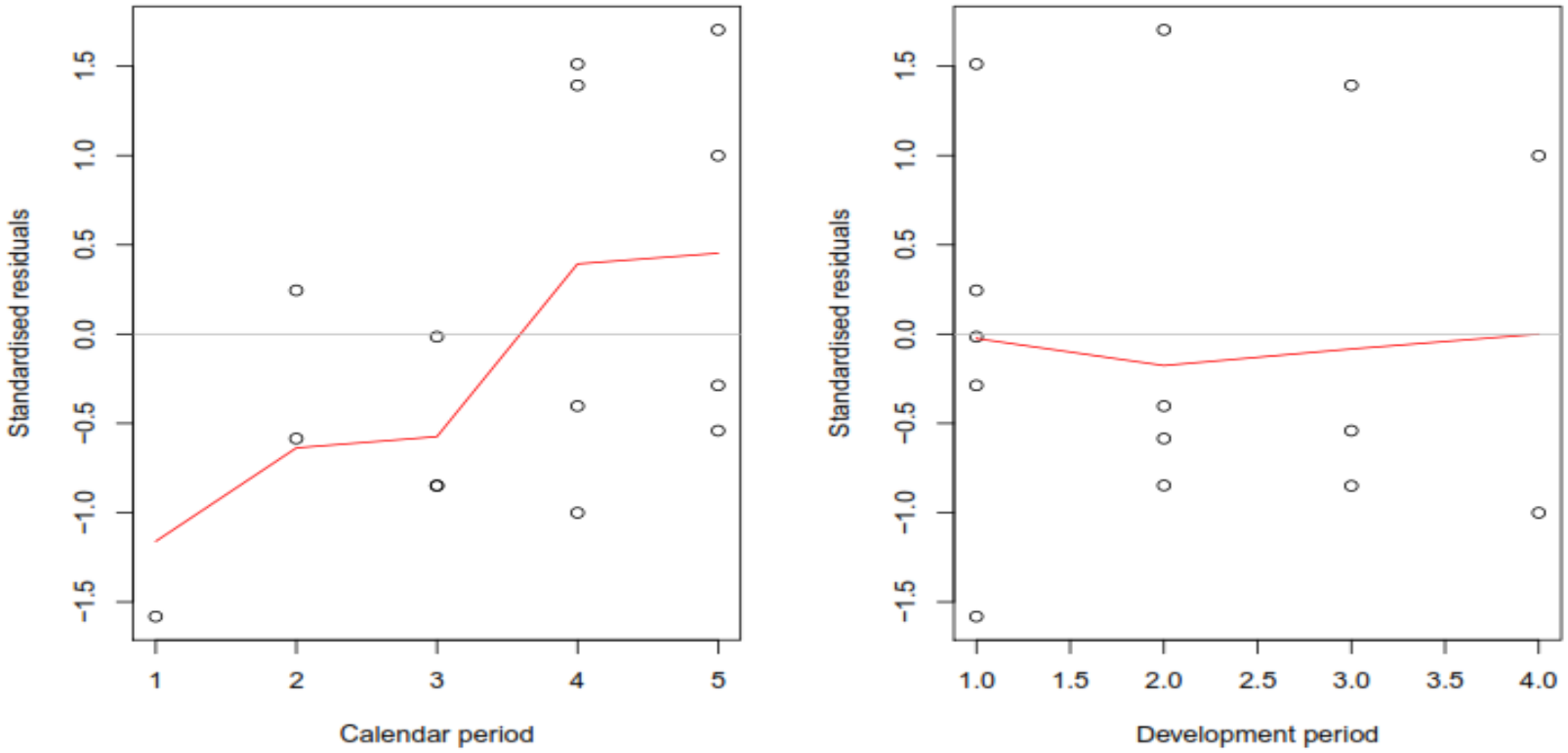


Figure 6 – Evolution of the standardized residuals of *j* and *i* + *j*

From Mack to Merz & Wüthrich

- We can plot the development, including the forecast and estimated standard errors by origin period by setting the argument `lattice=TRUE`.

```
> mack <- MackChainLadder(PAID)
> plot(mack, lattice=TRUE)
```

From Mack to Merz & Wüthrich

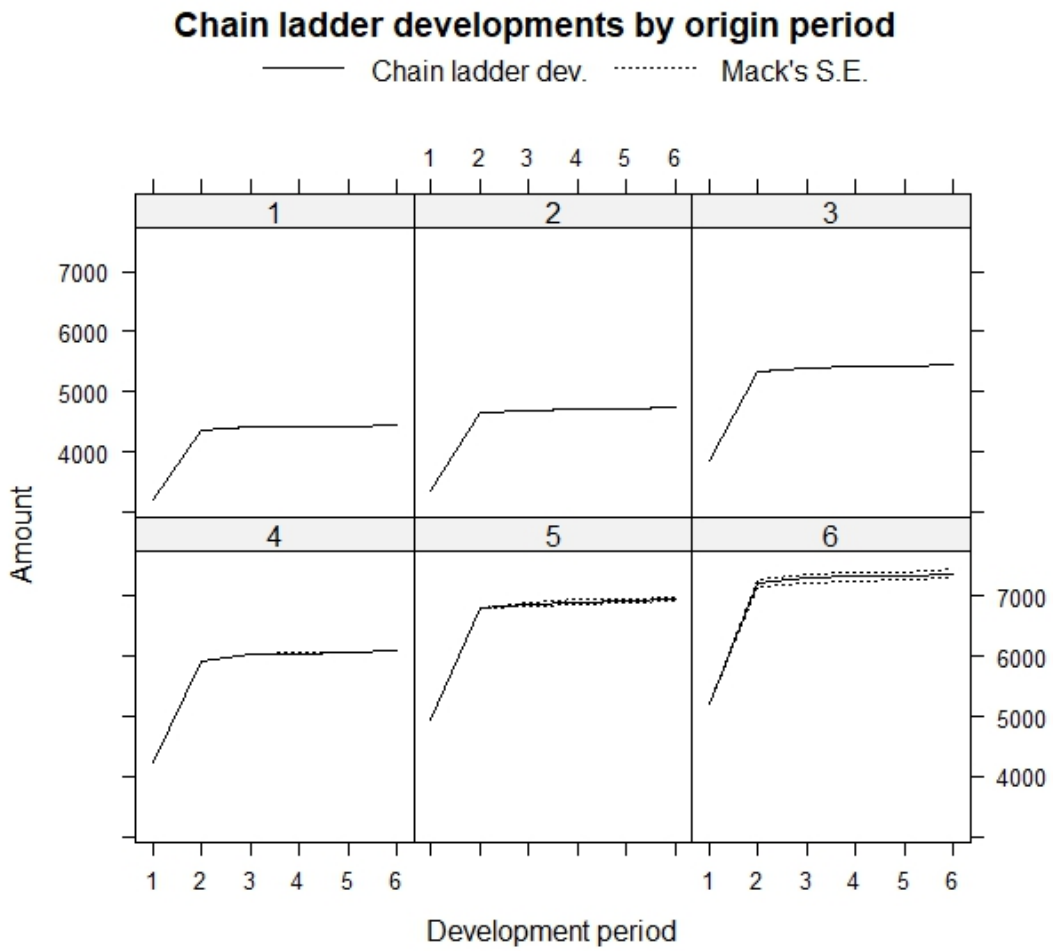


Figure 7 – Chain Ladder developments by orifin period

From Mack to Merz & Wüthrich

- The so-called *Munich Chain Ladder*, proposes using not only the cumulative payments, but also other information available: the estimated costs of the various claims made by claims handlers.

From Mack to Merz & Wüthrich

- Payments triangles were based on financial movements; these charges triangles are based on estimates made by management given information available to them. Late claims are not in, and certain claims will be filed without further action.

From Mack to Merz & Wüthrich

- However, it may appear legitimate to use this information. In the *Munich Chain Ladder* method, it has the triangles ($C_{i,j}$) corresponding to the cumulative payments, and ($\Gamma_{i,j}$) the charges say file / folder.

From Mack to Merz & Wüthrich

- We define ratios payments / charge, and charge / payment

$$\boldsymbol{Q_{i,j} = \frac{C_{i,j}}{\Gamma_{i,j}}}$$

- And

$$\boldsymbol{Q_{i,j}^{-1} = \frac{\Gamma_{i,j}}{C_{i,j}}}$$

From Mack to Merz & Wüthrich

- As in the basic chain-ladder model is assumed that

$$\begin{cases} \mathbb{E}\left(\boldsymbol{C}_{\mathbf{i,j+1}} | \boldsymbol{\mathcal{F}}^{\boldsymbol{C}}_{\mathbf{i+}\boldsymbol{j}}\right) = \boldsymbol{\lambda}^{\boldsymbol{C}}_{\boldsymbol{j}} \boldsymbol{C}_{\mathbf{i},\boldsymbol{j}} \\ \mathbb{E}\left(\boldsymbol{\Gamma}_{\mathbf{i,j+1}} | \boldsymbol{\mathcal{F}}^{\boldsymbol{\Gamma}}_{\mathbf{i+}\boldsymbol{j}}\right) = \boldsymbol{\lambda}^{\boldsymbol{\Gamma}}_{\boldsymbol{j}} \boldsymbol{\Gamma}_{\mathbf{i},\boldsymbol{j}} \end{cases}$$

- And

$$\begin{cases} \mathbb{V}\left(\boldsymbol{C}_{\mathbf{i,j+1}} | \boldsymbol{\mathcal{F}}^{\boldsymbol{C}}_{\mathbf{i+}\boldsymbol{j}}\right) = \left[\boldsymbol{\sigma}^{\boldsymbol{C}}_{\boldsymbol{j}}\right]^{\mathbf{2}} \boldsymbol{C}_{\mathbf{i},\boldsymbol{j}} \\ \mathbb{V}\left(\boldsymbol{\Gamma}_{\mathbf{i,j+1}} | \boldsymbol{\mathcal{F}}^{\boldsymbol{\Gamma}}_{\mathbf{i+}\boldsymbol{j}}\right) = \left[\boldsymbol{\sigma}^{\boldsymbol{\Gamma}}_{\boldsymbol{j}}\right]^{\mathbf{2}} \boldsymbol{\Gamma}_{\mathbf{i},\boldsymbol{j}} \end{cases}$$

From Mack to Merz & Wüthrich

- It also adds an information on $\lambda_{i,j}$. If

$$\lambda^{C}_{i,j-1} = \frac{C_{i,j}}{C_{i,j-1}}$$

- And

$$\lambda^{\Gamma}_{i,j-1} = \frac{\Gamma_{i,j}}{\Gamma_{i,j-1}}$$

From Mack to Merz & Wüthrich

- We assume that

$$\mathbb{E}\left(\boldsymbol{\lambda}_{i,j-1}^{C}|\boldsymbol{\mathcal{F}}_{i+j}\right)$$

$$= \boldsymbol{\lambda}_{j-1}^{C} + \boldsymbol{\lambda}^{C}\sqrt{\mathbb{V}\left(\boldsymbol{\lambda}_{i,j-1}^{C}|\boldsymbol{\mathcal{F}}_{i+j}^{C}\right)}.\frac{\boldsymbol{Q}_{i,j-1}^{-1} - \mathbb{E}\left(\boldsymbol{Q}_{i,j-1}^{-1}|\boldsymbol{\mathcal{F}}_{i+j^{C}}\right)}{\sqrt{\mathbb{V}\left(\boldsymbol{Q}_{i,j-1}^{-1}|\boldsymbol{\mathcal{F}}_{i+j^{C}}\right)}}$$

From Mack to Merz & Wüthrich

- And

$$\mathbb{E}\left(\boldsymbol{\lambda}_{i,j-1}^{\Gamma}|\boldsymbol{\mathcal{F}}_{i+j}\right)$$

$$= \boldsymbol{\lambda}_{j-1}^{\Gamma} + \boldsymbol{\lambda}^{\Gamma}\sqrt{\mathbb{V}\left(\boldsymbol{\lambda}_{i,j-1}^{\Gamma}|\boldsymbol{\mathcal{F}}_{i+j}^{\Gamma}\right)}.\frac{\boldsymbol{Q}_{i,j-1}^{-1} - \mathbb{E}\left(\boldsymbol{Q}_{i,j-1}^{-1}|\boldsymbol{\mathcal{F}}_{i+j^{\Gamma}}\right)}{\sqrt{\mathbb{V}\left(\boldsymbol{Q}_{i,j-1}^{-1}|\boldsymbol{\mathcal{F}}_{i+j^{\Gamma}}\right)}}$$

From Mack to Merz & Wüthrich

- It will be noted that this is an extension of the chain ladder model and in particular

$$\mathbb{E}\left(\boldsymbol{\lambda}_{\boldsymbol{i,j-1}}^{\boldsymbol{\Gamma}} | \boldsymbol{\mathcal{F}}_{\boldsymbol{i+j}}^{\boldsymbol{C}}\right) = \mathbb{E}\left[\mathbb{E}\left(\left(\boldsymbol{\lambda}_{\boldsymbol{i,j-1}}^{\boldsymbol{\Gamma}} | \boldsymbol{\mathcal{F}}_{\boldsymbol{i+j}}\right) | \boldsymbol{\mathcal{F}}_{\boldsymbol{i+j}}\right)^{\boldsymbol{C}}\right] = \boldsymbol{\lambda}_{\boldsymbol{j-1}}^{\boldsymbol{C}}$$

From Mack to Merz & Wüthrich

- The λ^C and λ^Γ terms are then simply conditional correlation coefficients. more precisely

$$\boldsymbol{\lambda^C = Cor\left(\Gamma_{i,j-1}, C_{i,j} | \mathcal{F}^C_{i+j-1}\right)}$$

From Mack to Merz & Wüthrich

- Under these assumptions, it is possible to construct estimators without bias

$$\mathbb{E}\left(\boldsymbol{C_{i,j}} | \boldsymbol{C_{i,j-1}}\right)$$

- And

$$\mathbb{E}\left(\boldsymbol{\Gamma_{i,j}} | \boldsymbol{\Gamma_{i,j-1}}\right)$$

From Mack to Merz & Wüthrich

- From respectively

$$\mathbb{E}\left(\boldsymbol{Q_{i,j}}|\boldsymbol{\mathcal{F}^{\Gamma}_{i+j}}\right)$$

- And of

$$\mathbb{E}\left(\boldsymbol{Q^{-1}_{i,j}}|\boldsymbol{\mathcal{F}^{C}_{i+j}}\right)$$

From Mack to Merz & Wüthrich

- To estimate the last two quantities, ask

$$\widehat{Q}_j = \frac{\sum_{i=0}^{n_j} C_{i,j}}{\sum_{i=0}^{n_j} \Gamma_{i,j}} = \frac{1}{\widehat{Q}_j^{-1}}$$

From Mack to Merz & Wüthrich

- We can also estimate the conditional variances. For example

$$\widehat{\mathbb{V}}\left(\boldsymbol{Q}_{i,j}|\boldsymbol{\mathcal{F}}_{i+j}^{\boldsymbol{\Gamma}}\right) = ()^{-1}\sum_{i=0}^{n-j}\boldsymbol{\Gamma}_{i,j}\left[\boldsymbol{Q}_{i,j} - \widehat{\boldsymbol{Q}}_{j}\right]^{2}$$

- And a similar expression for

$$\widehat{\mathbb{V}}\left(\boldsymbol{Q}_{i,j}^{-1}|\boldsymbol{\mathcal{F}}_{i+j}^{C}\right)$$

From Mack to Merz & Wüthrich

- From these amounts, then ask

$$\widetilde{Q}_{i,j} = \frac{Q_{i,j} - \widehat{Q}_j}{\sqrt{\widehat{\mathbb{V}}\left(Q_{i,j} | \mathcal{F}^{\Gamma}_{i+j}\right)}}$$

$$\tilde{\lambda}^{\Gamma}_{i,j} = \frac{\sqrt{\Gamma_{i,j-1}}}{\left[\widehat{\sigma}^{I}_{j-1}\right]^2 \left[\lambda_{i,j-1} - \hat{\lambda}_{j-1}\right]}$$

From Mack to Merz & Wüthrich

- And

$$\hat{\lambda}^{\Gamma} = \frac{\sum \widetilde{Q}_{i,j-1} \tilde{\lambda}^{\Gamma}_{i,j}}{\sum \widetilde{Q}^{2}_{i,j-1}}$$

- The Munich-Chain-Ladder estimator is built iteratively.

From Mack to Merz & Wüthrich

```
> INCURRED <-
matrix(c(4795,4629,4497,4470,4456,4456,5135,494
9,4783,4760,4750,NA,5681,5631,5492,5470,NA,NA,6
272,6198,6131,NA,NA,NA,7326,7087,NA,NA,NA,NA,73
53,NA,NA,NA,NA,NA), nrow = 6, ncol = 6, byrow =
T)
> INCURRED
```

From Mack to Merz & Wüthrich

	0	1	2	3	4	5
0	4795	4629	4497	4470	4456	4456
1	5135	4949	4783	4760	4750	
2	5681	5631	5492	5470		
3	6272	6198	6131			
4	7326	7087				
5	7353					

Table 7 - Triangle charges estimates for cumulative file / folder, $\Gamma = (\Gamma_{i,j})$

From Mack to Merz & Wüthrich

```
> (MNCL = MunichChainLadder(Paid = PAID,
Incurred = INCURRED))

Totals
            Paid Incurred P/I Ratio
Latest:   32,637   35,247      0.93
Ultimate: 35,238   35,230      1.00
```

From Mack to Merz & Wüthrich

	Latest Paid	Latest Incurred	Latest P/I Ratio	Ult. Paid	Ult. Incurred	Ult. P/I Ratio
1	4,456	4,456	1.000	4,456	4,456	1
2	4,731	4,751	0.996	4,753	4,751	1
3	5,421	5,471	0.991	5,455	5,454	1
4	6,021	6,131	0.982	6,086	6,084	1
5	6,794	7,087	0.959	6,974	6,972	1
6	5,217	7,353	0.710	7,514	7,513	1

From Mack to Merz & Wüthrich

- If we compare the two triangles, which were supplemented taking into account the interactions, we get relatively close things.

```
> plot(MNCL)
```

From Mack to Merz & Wüthrich

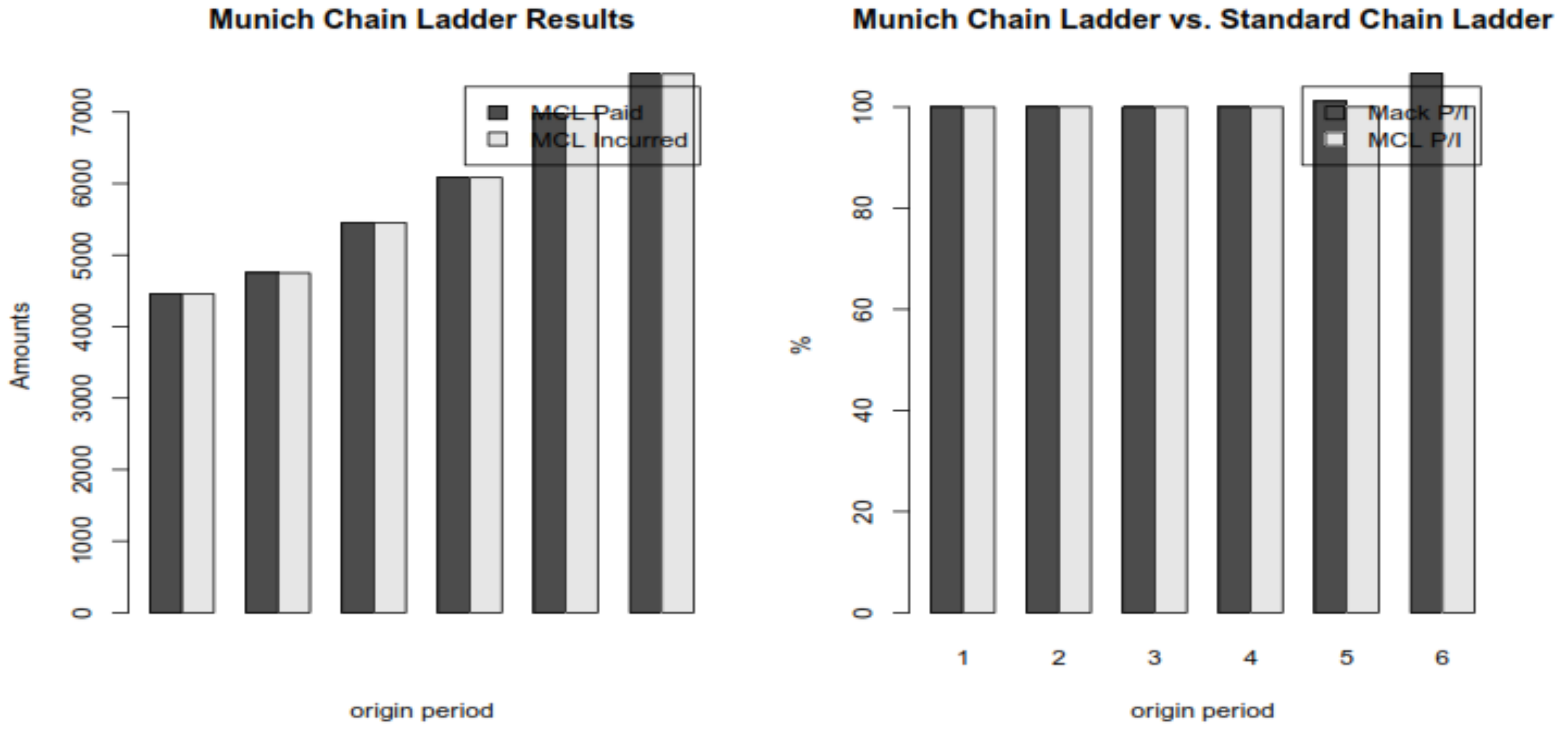


Figure 8 – Comparison of chain ladder methods, and Munich Chain Ladder, climbing to the left, and values on the right

From Mack to Merz & Wüthrich

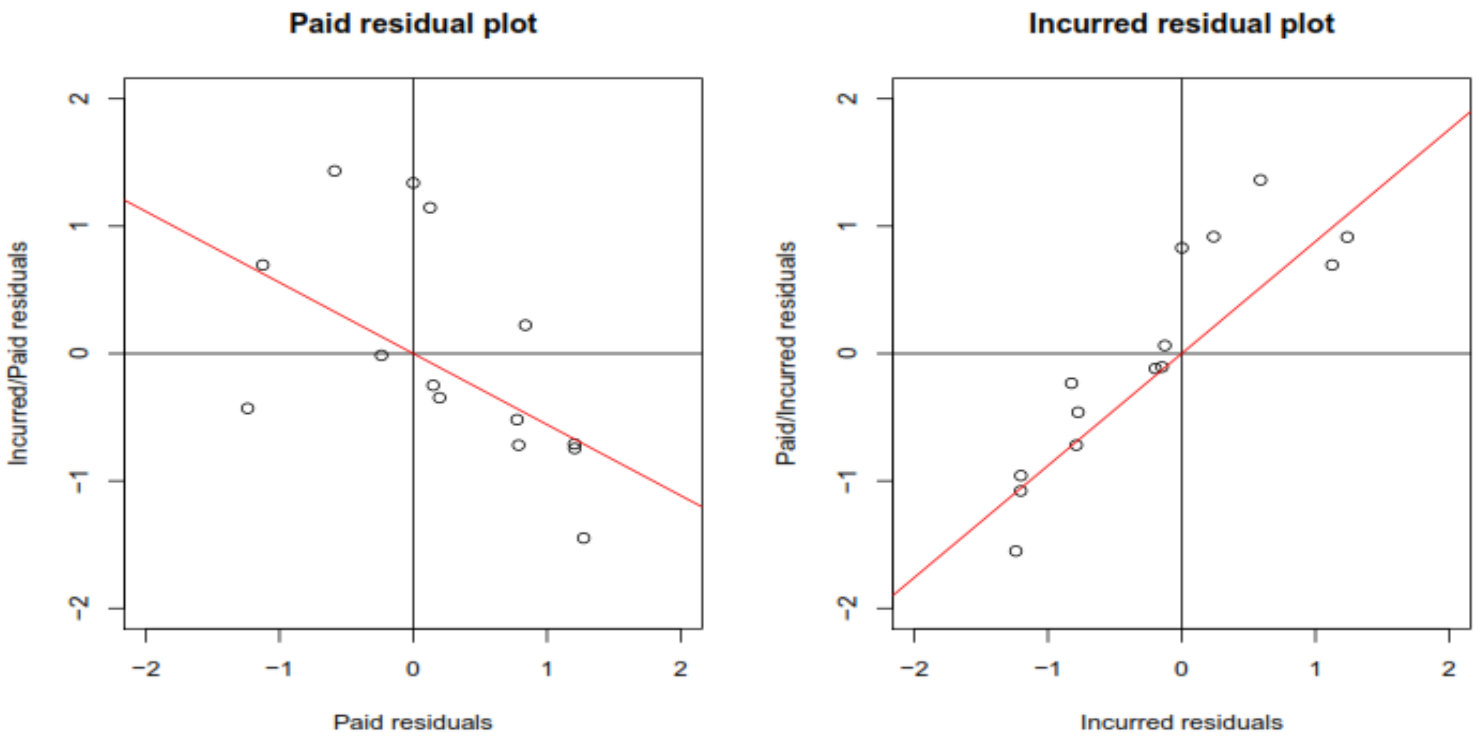


Figure 9 – Correlations between payments development triangles, and loads file / folder

From Mack to Merz & Wüthrich

```
> MNCL$MCLPaid
```

From Mack to Merz & Wüthrich

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]
[1,]	3209	4372.000	4411.000	4428.000	4435.000	4456.000
[2,]	3367	4659.000	4696.000	4720.000	4730.000	4752.532
[3,]	3871	5345.000	5398.000	5420.000	5429.793	5455.426
[4,]	4239	5917.000	6020.000	6046.102	6057.307	6085.918
[5,]	4929	6794.000	6886.091	6925.349	6941.120	6974.068
[6,]	5217	7244.985	7404.245	7457.998	7478.582	7514.281

From Mack to Merz & Wüthrich

```
> sum(MNCL$MCLPaid[,6] -
diag(MNCL$MCLPaid[,6:1]))
[1] 2601.227
> MNCL$MCLIncurred
```

From Mack to Merz & Wüthrich

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]
[1,]	4795	4629.000	4497.000	4470.000	4456.000	4456.000
[2,]	5135	4949.000	4783.000	4760.000	4750.000	4751.400
[3,]	5681	5631.000	5492.000	5470.000	5454.741	5453.947
[4,]	6272	6198.000	6131.000	6100.985	6085.006	6084.280
[5,]	7326	7087.000	6985.185	6978.864	6971.468	6972.317
[6,]	7353	7303.756	7476.573	7504.167	7509.575	7512.549

From Mack to Merz & Wüthrich

```
> sum(MNCL$MCLIncurred[,6] -
diag(MNCL$MCLPaid[,6:1]))
[1] 2593.492
```

From Mack to Merz & Wüthrich

- We can also use this technique to visualize the rates of payment, but also to estimate load file by file, as in Figure 9. We uses the function can be called for two years of different development, one year close to the left, and seen after 2 years of development right.

From Mack to Merz & Wüthrich

```
munich = function(k){
plot(0:(nc+1-k), c(0,MNCL$MCLPaid[k,1:(nc+1-
k)]), pch = 19, type = "b", ylim = c(0,
max(c(MNCL$MCLPaid[k,],
MNCL$MCLIncurred[k,]))), xlim = c(0,nc), ylab
="", xlab = "")
lines(0:(nc+1-k),
c(0,MNCL$MCLIncurred[k,1:(nc+1-k)]), pch = 19,
type = "b")
```

From Mack to Merz & Wüthrich

```
> lines((nc+1-k):nc,MNCL$MCLPaid[k,(nc+1-
k):nc], pch = 1, type = "b")
> lines((nc+1-k):nc,MNCL$MCLIncurred[k,(nc+1-
k):nc], pch = 1, type = "b")
> abline(v = nc+1-k, col = "grey")
}
> munich(5)
```

From Mack to Merz & Wüthrich

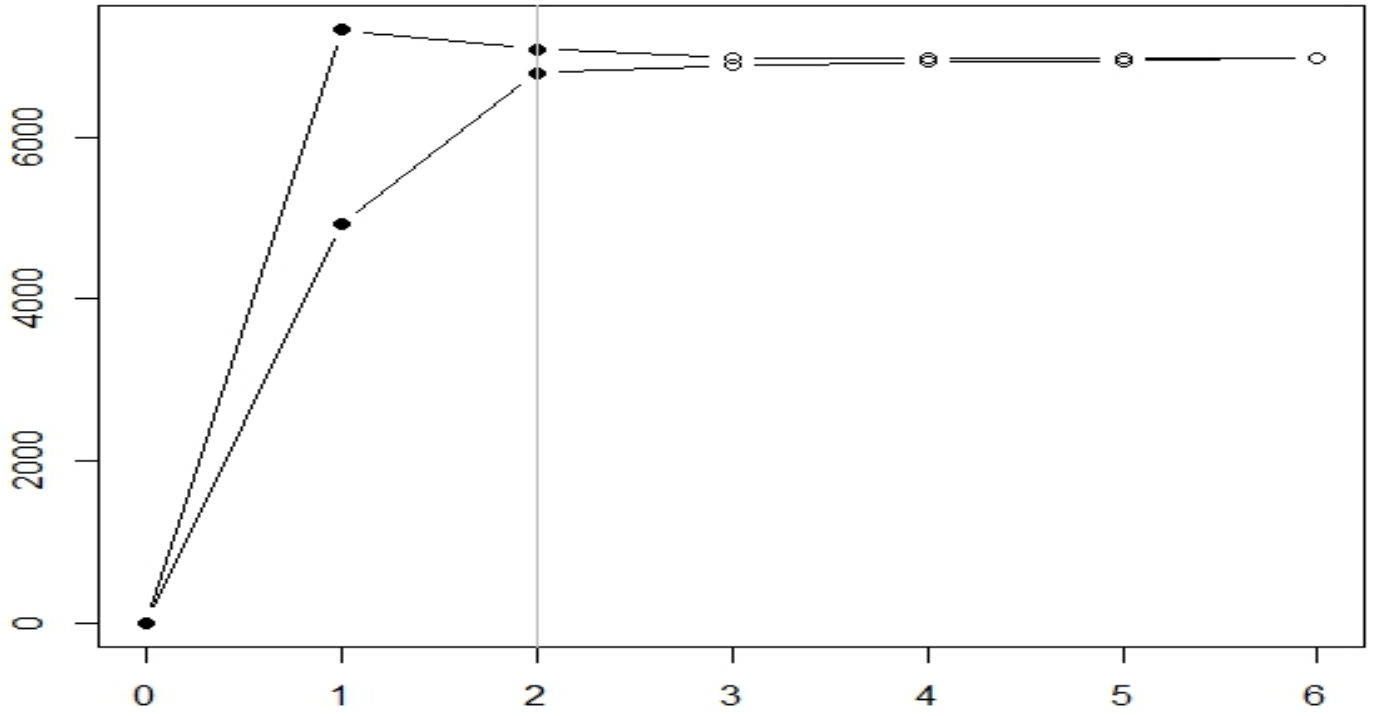


Figure 10 – Evolution of payments, and charge file by file

From Mack to Merz & Wüthrich

- Merz & Wüthrich studied the variation of the bonus / Mali from one year to the other (called CDRi(n) claims development result), that is to say the change in the prediction of the total load. In particular, they showed that

$$\widehat{\mathbf{msepc}}_{\mathbf{n-1}}(\mathbf{CDR_i(n)}) = \hat{\mathbf{C}}^{\mathbf{2}}_{\mathbf{i},\infty}\left(\hat{\boldsymbol{\Gamma}}_{\mathbf{i,n}} + \widehat{\boldsymbol{\Delta}}_{\mathbf{i,n}}\right)$$

From Mack to Merz & Wüthrich

- Where

$$\hat{\Delta}_{i,n} = \frac{\hat{\sigma}^2_{n-i+1}}{\hat{\lambda}^2_{n-i+1} S^{n+1}_{n-i+1}} + \sum_{j=n-i+2}^{n-1} \left(\frac{C_{n-j+1,j}}{S^{n+1}_j} \right)^2 \frac{\hat{\sigma}^2_j}{\hat{\lambda}^2_j S^n_j}$$

From Mack to Merz & Wüthrich

- And

$$\widehat{\Gamma}_{i,n} = \left(1 + \frac{\widehat{\sigma}^2_{n-i+1}}{\widehat{\lambda}^2_{n-i+1} C_{i,n-i+1}}\right) \prod_{j=n-i+2}^{n-1} \left(1 + \frac{\widehat{\sigma}^2_j}{\widehat{\lambda}^2_j \left[S^{n+1}_j\right]^2} C_{n-j+1,j}\right) - 1$$

From Mack to Merz & Wüthrich

- Which is approached by

$$\widehat{\Gamma}_{i,n} \approx \frac{\widehat{\sigma}^2_{n-i+1}}{\widehat{\lambda}^2_{n-i+1} C_{i,n-i+1}} + \sum_{j=n-i+2}^{n-1} \left(\frac{C_{n-j+1,j}}{S_j^{n+1}} \right)^2 \frac{\widehat{\sigma}^2_j}{\widehat{\lambda}^2_j C_{n-j+1,j}}$$

From Mack to Merz & Wüthrich

- By simply making a development of the form

$$\prod(1+u_i) \approx 1 + \sum u_i$$

From Mack to Merz & Wüthrich

- But that is only valid if u_i is small, here

$$\frac{\hat{\sigma}_j^2}{\hat{\lambda}_j^2} \ll C_{n-j+1,j}$$

From Mack to Merz & Wüthrich

```
> M <- MackChainLadder(PAID, est.sigma="Mack")
> cdrM <- CDR(M)
> round(cdrM, 1)
```

From Mack to Merz & Wüthrich

	IBNR	CDR(1)S.E.	Mack.S.E.
1	0.0	0.0	0.0
2	22.4	1.4	1.4
3	35.8	2.5	2.9
4	66.1	4.5	5.3
5	153.1	30.9	31.4
6	2149.7	60.8	68.5
Total	2427.0	72.6	79.5

Poissonian regression and econometric approaches

- In this part, we will move away recursive models inspired by the chain ladder method, and we will return to the model classes widely used in the 70s, called factor models, brought up to date by offering a playback of these econometric models, thus providing confidence intervals of different sizes.

Poissonian regression and econometric approaches

- Before introducing the use of regression models, we can begin by recalling older models. For example Taylor assumed that

$$Y_{i,j} = r_j \times \mu_{i+j}$$

- For any *i,j*
- i.e. a column effect rate of payment, and a diagonal effect that Taylor interprets as an inflation factor.

Poissonian regression and econometric approaches

- This model can be rewritten, since there is no positive increment,

$$\boldsymbol{logY_{i,j} = \alpha_i + \gamma_{i+j}}$$

- Who takes a linear form. Later we will show that if

$$\boldsymbol{logY_{i,j} = \alpha_i + \beta_j}$$

- Is like a *Chain-Ladder* model type.

Poissonian regression and econometric approaches

- Indeed, this assumes that

$$\boldsymbol{Y_{i,j} = \alpha_i \times \beta_j}$$

- That can approach the development model

$$\boldsymbol{Y_{i,j} = C_{i,n} \times \varphi_j}$$

Poissonian regression and econometric approaches

- De Vylder was one of the first econometric models provisioning. It is assumed that

$$\boldsymbol{Y_{i,j} \sim \mathcal{N}(\alpha_i \times \beta_j, \sigma^2)}$$

- For any i,j. We can estimate the coefficients by least squares,

$$\boldsymbol{(\widehat{\alpha}, \widehat{\beta}) = argmin\left\{\sum_{i,j}\left[Y_{i,j} - \alpha_i \times \beta_j\right]^2\right\}}$$

Poissonian regression and econometric approaches

- The normal equations are written here

$$\widehat{\boldsymbol{\alpha}}_{\boldsymbol{i}} = \frac{\sum_{\boldsymbol{j}} \boldsymbol{Y}_{\boldsymbol{i},\boldsymbol{j}} \times \widehat{\boldsymbol{\beta}}_{\boldsymbol{j}}}{\sum_{\boldsymbol{j}} \widehat{\boldsymbol{\beta}}_{\boldsymbol{j}}^{\boldsymbol{2}}}$$

- And

$$\widehat{\boldsymbol{\beta}}_{\boldsymbol{i}} = \frac{\sum_{\boldsymbol{j}} \boldsymbol{Y}_{\boldsymbol{i},\boldsymbol{j}} \times \widehat{\boldsymbol{\alpha}}_{\boldsymbol{j}}}{\sum_{\boldsymbol{j}} \widehat{\boldsymbol{\alpha}}_{\boldsymbol{j}}^{\boldsymbol{2}}}$$

Poissonian regression and econometric approaches

- What is not explicitly resolved. To solve it, Christofides suggested to rewrite it as a log-linear model, i.e.

$$\boldsymbol{logY_{i,j} \sim \mathcal{N}(a_i + b_j, \sigma^2)}$$

- For any i,j.

Poissonian regression and econometric approaches

```
ligne <- rep(1:nl, each = nc); colonne <-
rep(1:nc, nl)
INC <- PAID
INC[,2:6] <- PAID[,2:6] - PAID[,1:5]
Y <- as.vector(INC)
lig <- as.factor(ligne)
col <- as.factor(colonne)
reg <- lm(log(Y) ~ col + lig)
summary(reg)
```

Poissonian regression and econometric approaches

- Then we can simply use this regression to construct the base triangle of the model. As we noted earlier, you cannot use

$$\widehat{Y}_{i,j} = exp\left[\widehat{a}_i + \widehat{b}_j\right]$$

- For this estimator is biased, however.

Poissonian regression and econometric approaches

- If the bias is corrected by

$$\widehat{Y}_{i,j} = exp\left[\widehat{a}_i + \widehat{b}_j + \frac{\widehat{\sigma}^2}{2}\right]$$

- Because

$$exp\left(\mathbb{E}\left(log(Y)\right)\right) \neq \mathbb{E}(Y))$$

Poissonian regression and econometric approaches

- Is then obtained

```
sigma <- summary(reg)$sigma
(INCpred <- matrix(exp(log(Y) + sigma^2/2), nl,
nc))
sum(exp(log(Y)[is.na(Y) == TRUE] + sigma^2/2))
```

Poissonian regression and econometric approaches

- *Hachemeister & Stanard* showed that in a *log-Poisson* regression on the increments, the amount of future payments predictions corresponds to the estimator Chain Ladder.

Poissonian regression and econometric approaches

```
CL <- glm(Y ~ lig + col, family = poisson)
summary(CL)
```

- Note now that the Poisson model is not necessarily the most suitable. Indeed, there is a strong overdispersion payments.

```
CL <- glm(Y ~ lig + col, family = quasipoisson)
summary(CL)
```

Poissonian regression and econometric approaches

- There is here a 2n - 1 parameters to be estimated, γ, c = (c1, ..., cn-1) and r = (r1, ..., rn-1), excluding the over-dispersion parameter Phi.

Poissonian regression and econometric approaches

- Given the variety of factors, here a row r and a column factor c, once these estimated parameters, it is possible to predict the bottom of the triangle very simply, i.e.

$$\widehat{Y}_{i,j} = \widehat{\mu}_{i,j} = exp\left[\widehat{\gamma} + \widehat{r}_i + \widehat{c}_j\right]$$

Poissonian regression and econometric approaches

```
Ypred <- predict(CL, newdata =
data.frame(lig,col), type = "response")
(INCpred <- matrix(Ypred, nl, nc))
```

- Here we find the estimator obtained by the Chain-Ladder method.

```
sum(Ypred[is.na(Y) == TRUE])
```

Poissonian regression and econometric approaches

- The reference value is the value in the upper left corner. Given the logarithmic form of the model, has a simple interpretation of all values, with respect to this first value

$$\mathbb{E}\left(\boldsymbol{Y_{i,j}}|\boldsymbol{\mathcal{F}_n}\right) = \mathbb{E}\left(\boldsymbol{Y_{0,0}}|\boldsymbol{\mathcal{F}_n}\right) \times \boldsymbol{exp}[\boldsymbol{r_i} + \boldsymbol{c_i}]$$

Poissonian regression and econometric approaches

- In some triangles, it is not uncommon to have negative increments. For example, consider the following payments triangle

Poissonian regression and econometric approaches

```
PAIDneg <-
matrix(c(3209,4372,4411,4428,4435,4456,3367,465
9,4696,4720,4730,NA,3871,5345,5338,5420,NA,NA,4
239,5917,6020,NA,NA,NA,4929,6794,NA,NA,NA,NA,52
17,NA,NA,NA,NA,NA), nrow = 6, ncol = 6, byrow =
T)
PAIDneg
```

Poissonian regression and econometric approaches

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]
[1,]	3209	4372	4411	4428	4435	4456
[2,]	3367	4659	4696	4720	4730	NA
[3,]	3871	5345	5338	5420	NA	NA
[4,]	4239	5917	6020	NA	NA	NA
[5,]	4929	6794	NA	NA	NA	NA
[6,]	5217	NA	NA	NA	NA	NA

Poissonian regression and econometric approaches

```
INCneg = PAIDneg
INCneg[,2:6] = PAIDneg[,2:6] -
PAIDneg[,1:5]
INCneg
```

Poissonian regression and econometric approaches

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]
[1,]	3209	1163	39	17	7	21
[2,]	3367	1292	37	24	10	NA
[3,]	3871	1474	-7	82	NA	NA
[4,]	4239	1678	103	NA	NA	NA
[5,]	4929	1865	NA	NA	NA	NA
[6,]	5217	NA	NA	NA	NA	NA

Poissonian regression and econometric approaches

- This negative increment payment in no way disturbs the Chain-Ladder method.

```
LAMBDAneg <- rep(NA, nc - 1)
for(k in 1:(nc - 1)){
LAMBDAneg[k] = lm(PAIDneg[,k+1] ~ 0 +
PAIDneg[,k], weights = 1 /
PAIDneg[,k])$coefficients}
TRIANGLEneg <- PAIDneg
```

Poissonian regression and econometric approaches

```
for(i in 1:(nc-1)){
TRIANGLEneg[(nl-i+1):(nl),i+1] = LAMBDAneg[i] *
TRIANGLEneg[(nl-i+1):(nl),i]}
chargeultimeneg <- TRIANGLEneg[,nc]
paiementsneg <- diag(TRIANGLEneg[,nc:1])
RESERVESneg <- chargeultimeneg - paiementsneg
sum(RESERVESneg)
[1] 2469.703
```

Poissonian regression and econometric approaches

- However, both methods of regression that we just presented cannot be applied. If however you want to use this technique, a first solution is to rebalance payments from one year to another. One can then take a left or right so as to no longer have this payment negative increment.

Poissonian regression and econometric approaches

- Another strategy can be to make translations for certain increments. If we translate all observations, the code would look like

```
predict(glm((Y + k) ~ X); type = “response”) –
k
```

Poissonian regression and econometric approaches

- Indeed, in a linear Gaussian model, translating the observations Y up then translate down the $\hat{Y}$ predictions of the same amount given exactly the same predictions. But this is not the case in the GLM models.

Poissonian regression and econometric approaches

- Suppose we translate the increments of the column where represent the negative increment, so that all the increments are then positive. We can then run a Poisson regression. By translating so as to cancel the negative increment, we get the following provision amount

Poissonian regression and econometric approaches

```
translation <- function(k){
Y = as.vector(INCneg)
Y[col == 3] = Y[col == 3] + k
base <- data.frame(Y, lig, col)
reg <- glm(Y ~ lig + col, data = base, family =
poisson(link = "log"))
```

Poissonian regression and econometric approaches

```
Yp = predict(reg, type = "response", newdata =
base)
Yp[col == 3] = Yp[col == 3] - k
sum(Yp[is.na(Y) == TRUE])}
translation(7)
```

```
[1] 2471.444
```

Poissonian regression and econometric approaches

- A solution can then be translated to a number of values, and then to extrapolate prediction zero for k.

```
K <- 7:20
reserves <- Vectorize(translation)(K)
(pRes <- predict(lm(reserves ~ K), newdata = (K
= 0)))
plot(reserves, xlab = "Translation", ylab =
"Poisson")
```

Poissonian regression and econometric approaches

- We can also see this extrapolation in the figure below.

```
plot(reserves, xlab = "Translation", ylab =
"Poisson")
```

```
[1] 2469.752
```

Poissonian regression and econometric approaches

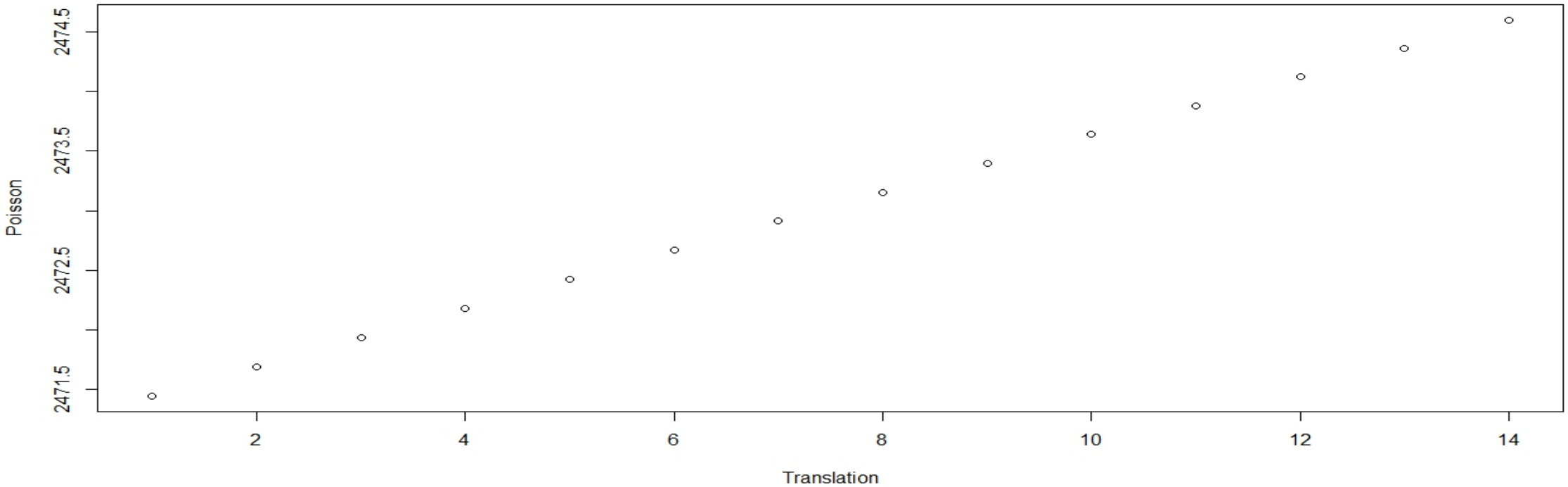


Figure 11 – Extrapolation of the amount of provision on a Poissonian regression with a negative increment (with translations).

Uncertainty in a regression model

- We noted earlier that getting an estimate of the amount of claims remaining to be paid was not enough, and had to have a confidence interval, or at least a measure of the dispersion of the true value around this predicted value, even a quantile.

Uncertainty in a regression model

- The regression models might look very interesting because there are closed formulas for all kinds of prediction. For example, in a GLM regression with a logarithmic link, remember that

$$\mathbb{E}\left(\boldsymbol{Y_{i,j}}|\boldsymbol{\mathcal{F}_n}\right) = \boldsymbol{\mu_{i,j}} = \boldsymbol{exp}\left[\boldsymbol{\eta_{i,j}}\right]$$

- or again

$$\widehat{\boldsymbol{Y}}_{\boldsymbol{i,j}} = \widehat{\boldsymbol{\mu}}_{\boldsymbol{i,j}} = \boldsymbol{exp}\left[\widehat{\boldsymbol{\eta}}_{\boldsymbol{i,j}}\right]$$

Uncertainty in a regression model

- The delta method that allows us to write

$$\mathbb{V}\left(\widehat{\boldsymbol{Y}}_{\boldsymbol{i},\boldsymbol{j}}\right) \approx \left|\frac{\boldsymbol{\partial \mu_{i,j}}}{\boldsymbol{\partial \eta_{i,j}}}\right|^{\mathbf{2}} \times \mathbb{V}\left(\widehat{\boldsymbol{\eta}}_{\boldsymbol{i},\boldsymbol{j}}\right)$$

- Which is simplified in the case where the link is logarithmic, i.e.

$$\frac{\boldsymbol{\partial \mu_{i,j}}}{\boldsymbol{\partial \eta_{i,j}}} = \boldsymbol{\mu_{i,j}}$$

Uncertainty in a regression model

- Also, for a Poisson overdispersed

$$\mathbb{E}\left(\left[\boldsymbol{Y}_{i,j} - \widehat{\boldsymbol{Y}}_{i,j}\right]^{\mathbf{2}}\right) \approx \widehat{\boldsymbol{\phi}} \times \widehat{\boldsymbol{\mu}}_{i,j} + \widehat{\boldsymbol{\mu}}_{i,j}^{\mathbf{2}} \times \widehat{\mathbb{V}}\left(\widehat{\boldsymbol{\eta}}_{i,j}\right)$$

- For the lower part of the triangle.

Uncertainty in a regression model

- Moreover, because it will be necessary to sum all the terms of the bottom of the triangle to determine the total amount of provisions,

$$\boldsymbol{Cov}\left(\widehat{\boldsymbol{Y}}_{\boldsymbol{i,j}}, \widehat{\boldsymbol{Y}}_{\boldsymbol{k,l}}\right) \approx \widehat{\boldsymbol{\mu}}_{\boldsymbol{i,j}} \times \widehat{\boldsymbol{\mu}}_{\boldsymbol{k,l}} \times \widehat{\boldsymbol{Cov}}\left(\widehat{\boldsymbol{\eta}}_{\boldsymbol{i,j}}, \widehat{\boldsymbol{\eta}}_{\boldsymbol{k,l}}\right)$$

Uncertainty in a regression model

- The amount of the provision to be estimated is the sum of future payments predictions,

$$\widehat{R} = \sum_{i+j>n} \widehat{Y}_{i,j}$$

Uncertainty in a regression model

- Then

$$\mathbb{E}\left(\left[\boldsymbol{R}-\widehat{\boldsymbol{R}}\right]^{\mathbf{2}}\right) \approx \left(\sum_{\boldsymbol{i+j>n}} \widehat{\boldsymbol{\phi}} \times \widehat{\boldsymbol{\mu}}_{\boldsymbol{i,j}}\right) + \widehat{\boldsymbol{\mu}}'_{\boldsymbol{F}} \times \widehat{\mathbb{V}}(\widehat{\boldsymbol{\eta}}_{\boldsymbol{F}}) \times \widehat{\boldsymbol{\mu}}_{\boldsymbol{F}}$$

- Where the vectors $\hat{\mu}_F$ and $\hat{\eta}_F$ are restrictions of $\hat{\mu}$ and $\hat{\eta}$ vectors which indices i + j > n (i.e. at the bottom of the triangle to predict).

Uncertainty in a regression model

```
p <- nl + nc - 1
phi <- sum(residuals(CL, "pearson")^2) / (sum(is.na(Y) == FALSE) - p)
Sig <- vcov(CL)
X <- model.matrix(glm(Ypred ~ lig + col, family = quasipoisson))
```

Uncertainty in a regression model

```
Cov.eta <- X%*%Sig%*%t(X)

Ypred <- predict(CL, newdata = data.frame(lig,col), type = "response")*(is.na(Y) == TRUE)

se2 <- phi * sum(Ypred) + t(Ypred) %*% Cov.eta %*% Ypred

sqrt(se2)
```

Uncertainty in a regression model

- The simulation methods are a good alternative if we have too little data to invoke asymptotic theorems. Remember that there are two sources of uncertainty
  - Process error
  - Variance error
- It will then be necessary to use two algorithms to quantify these errors.

Uncertainty in a regression model

- To quantify the estimation error, it is natural to simulate fake triangles (above) and then watch the distribution of estimators amount of provisions obtained for each triangle. Then, it generates a pseudo triangle using Pearson residues. Remember that for a Poisson regression,

$$\hat{\varepsilon}_{i,j} = \frac{Y_{i,j} - \widehat{Y}_{i,j}}{\sqrt{\widehat{Y}_{i,j}}}$$

Uncertainty in a regression model

- However, these residues need to be adjusted to have unit variance. Then it is considered conventionally

$$\hat{\varepsilon}_{i,j} = \sqrt{\frac{n}{n-k}} \times \frac{Y_{i,j} - \hat{Y}_{i,j}}{\sqrt{\hat{Y}_{i,j}}}$$

- Where k is the number of parameters estimated in the model.

Uncertainty in a regression model

```
(residus = residuals(CL, type = "pearson"))
n = sum(is.na(Y) == FALSE)
k = ncol(PAID) + nrow(PAID) - 1
(residus = sqrt(n / (n - k)) * residus)
epsilon <- rep(NA, nl * nc)
epsilon[is.na(Y) == FALSE] = residus
matrix(epsilon, nl, nc)
```

Uncertainty in a regression model

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]
[1,]	1.374976e+00	-1.63464607	-2.221573	-0.7100321	-0.6196386	3.3704e-15
[2,]	3.485024e-02	0.40189396	-3.207593	1.1490494	0.6000048	NA
[3,]	1.693203e-01	0.08216186	-1.484151	-0.430738	NA	NA
[4,]	-1.569329e+00	1.29257769	6.140566	NA	NA	NA
[5,]	1.887862e-01	-0.30587642	NA	NA	NA	NA
[6,]	3.649467e-14	NA	NA	NA	NA	NA

Uncertainty in a regression model

- By simulating errors (which are assumed to be independent and identically distributed),

$$\tilde{\varepsilon}^{b} = \left(\tilde{\varepsilon}_{i,j}^{b}\right)$$

- We then set

$$Y_{i,j}^{b} = \widehat{Y}_{i,j} + \sqrt{\widehat{Y}_{i,j}} \times \tilde{\varepsilon}_{i,j}^{b}$$

Uncertainty in a regression model

- To generate errors, the most common method is to use a non-parametric simulation, that is to say that we will bootstrap residues among the pseudo-residuals obtained.

Uncertainty in a regression model

- Otherwise it is also possible to use a parametric model. The distribution of residuals can be achieved by the following code, and visualized in Figure below.

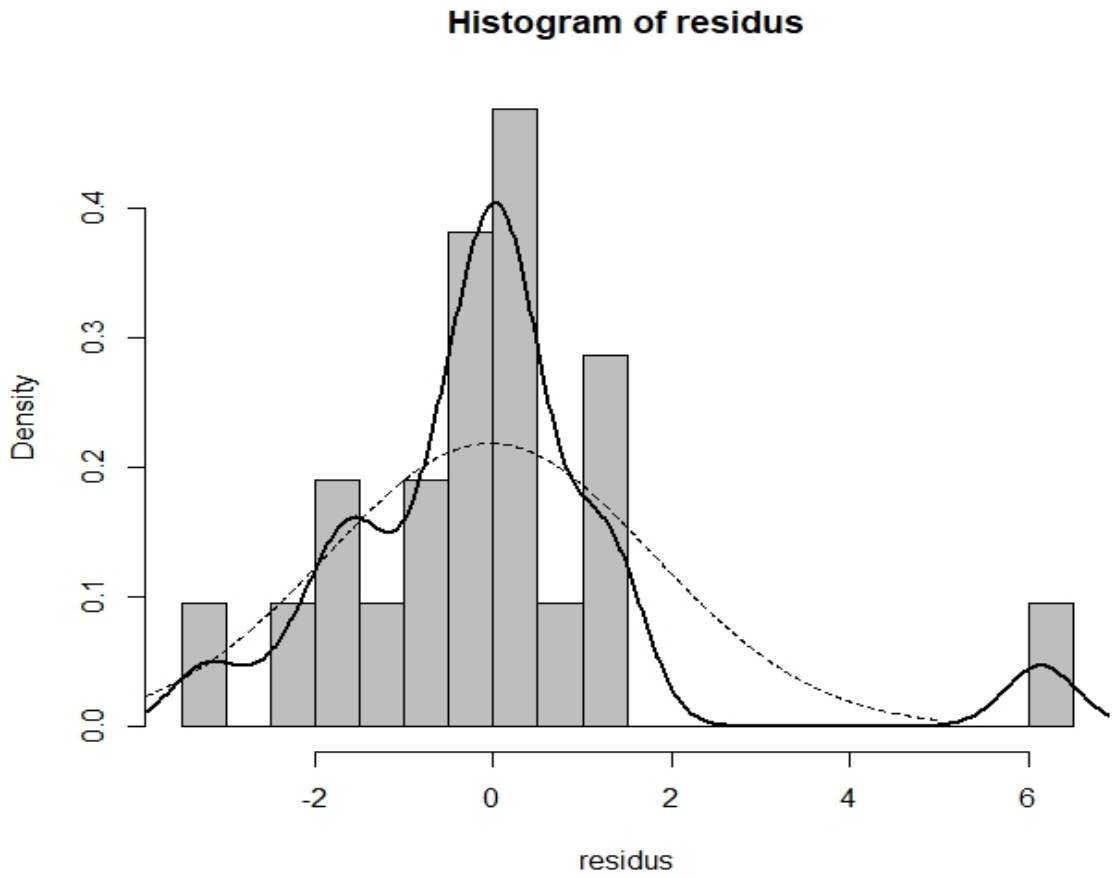
Uncertainty in a regression model

```
par(mfrow = c(1, 2))
hist(residus, breaks = seq(-3.5, 6.5, by = .5),
col = "grey", proba = TRUE)
u <- seq(-4, 5, by = .01)
densite <- density(residus)
lines(densite, lwd = 2)
lines(u, dnorm(u, mean(residus), sd(residus)),
lty = 2)
```

Uncertainty in a regression model

```
plot(ecdf(residus), xlab = "residus", ylab =
"Fn(residus)")
lines(u, pnorm(u, mean(residus), sd(residus)),
lty = 2)
Femp <- cumsum(densite$y) / sum(densite$y)
lines(densite$x, Femp, lwd = 2)
par(mfrow = c(1, 1))
```

Uncertainty in a regression model



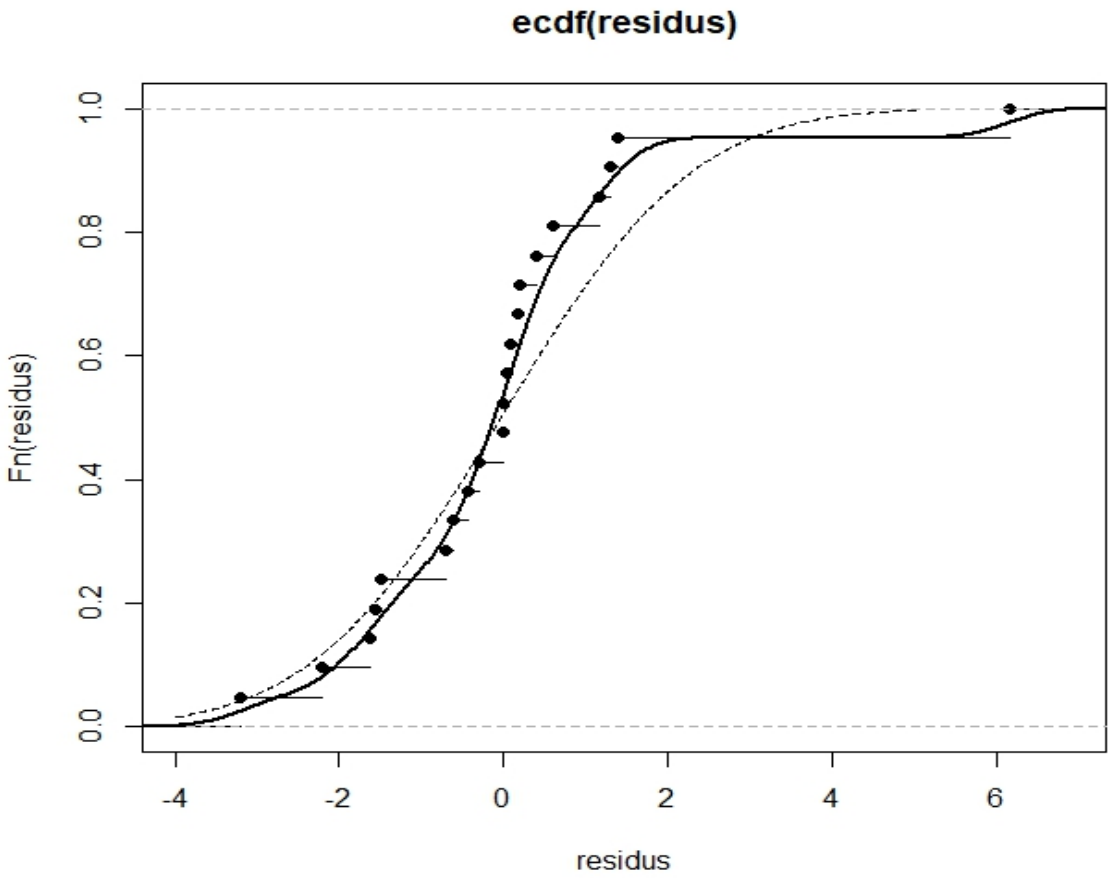


Figure 12 - Histogram and density residues (left) and distribution functions (right), with the Gaussian adjustment dotted.

Uncertainty in a regression model

- Once simulated pseudo-triangle payments increments, it is predicted the amount of provision $\hat{R}^b$. The variance of $\hat{R}^b$ is the estimation error.

Uncertainty in a regression model

- To take account of the model error, several methods may be used. The first, and simplest, is to note that from the pseudo triangle $Y_{i,j}^{b}$, we can get predictions for the lower part $\hat{Y}_{i,j}^{b}$.

Uncertainty in a regression model

- Given the Poissonian model, we can then simulate a trajectory for payments increments simulating $Y_{i,j}^{b}$ using the Poisson parameter $\hat{Y}_{i,j}^{b}$. The code is as follows

```
CLsimul1 <- function(triangle){
rpoisson(length(triangle), lambda = triangle)}
```

Uncertainty in a regression model

- However, simulating Poisson distributions may be too conservative. Indeed, as we saw on the quasiPoisson regression, over-dispersion parameter ϕ is significant.

```
(summary(CL)$dispersion)
```

Uncertainty in a regression model

- It may then be appropriate to generate laws more variance. The first idea may be to use a gamma distribution. Noting that $\mathbb{E}(Y) = \lambda$ and $\mathbb{V}(Y) = \phi\lambda$, the law of Gamma $G(\alpha,\beta)$ check $\alpha\beta = \lambda$ and $\alpha\beta 2 = \phi\lambda$. Also be used

Uncertainty in a regression model

```
rqpoisG <- function(n, lambda, phi, roundvalue
= TRUE) {
b <- phi
a <- lambda / phi
r <- rgamma(n, shape = a, scale = b)
if(roundvalue) {r <- round(r)}
return(r)}
```

Uncertainty in a regression model

- Another idea may be to use the relationship between the Poisson and negative binomial. For a negative binomial distribution with mean μ and variance $\mu + \mu 2 / k$, we set $\mu = \lambda$ and $k = \lambda (\phi\lambda - 1) -1$, i.e.

Uncertainty in a regression model

```
rqpoisBN = function(n, lambda, phi) {
mu <- lambda
k <- mu / (phi * mu -1)
r <- rnbinom(n, mu = mu, size = k)
return(r)}
```

Uncertainty in a regression model

- We then use the following code to generate payments scenarios,

```
CLsimul2 <- function(triangle, surdispersion){
rqpoissonG(length(triangle), lambda = triangle,
phi = surdispersion)}
```

Uncertainty in a regression model

- The second method is to use a replay Mack model proposed by England and Verrall. From the pseudo triangle, we will use the development factors and variances $\hat{\lambda}_j$ associated $\hat{\sigma}_j^2$ obtained on the initial triangle. It then extends the triangle in the lower part via the dynamic model

$$\widehat{\boldsymbol{C}}_{\boldsymbol{i},\boldsymbol{j}+\mathbf{1}}^{\boldsymbol{b}} | \widehat{\boldsymbol{C}}_{\boldsymbol{i},\boldsymbol{j}}^{\boldsymbol{b}} \sim \mathcal{N}\left(\hat{\boldsymbol{\lambda}}_{\boldsymbol{j}} \widehat{\boldsymbol{C}}_{\boldsymbol{i},\boldsymbol{j}}^{\boldsymbol{b}}, \widehat{\boldsymbol{\sigma}}_{\boldsymbol{j}}^{\mathbf{2}} \widehat{\boldsymbol{C}}_{\boldsymbol{i},\boldsymbol{j}}^{\boldsymbol{b}}\right)$$

Uncertainty in a regression model

- The code is the following, where triangle is a triangle cumulative payments (rather than increments in vector form), l corresponds to a vector of development factors and s to a volatilities vector,

Uncertainty in a regression model

```
CLsimul3 <- function(triangle, l, s){
m <- nrow(triangle)
for(i in 2:m){
triangle[(m-i+2):m,i]  <-  rnorm(i-1,  mean  =
triangle[(m-i+2):m,i-1]*l[i-1],
sd = sqrt(triangle[(m-i+2):m,i-1])*s[i-1])
}
return(triangle)}
```

Uncertainty in a regression model

- The overall algorithm to generate final expense estimates, but also scenarios of future payments is as follows

```
base <- data.frame(Y, lig, col)
ns <- 20000
set.seed(1)
Yp <- predict(CL, type = "response", newdata =
base)
Rs <- R <- rep(NA, ns)
for(s in 1:ns){
```

Uncertainty in a regression model

```
serreur <- sample(residus, size = nl*nc,
replace = TRUE)
E <- matrix(serreur, nl, nc)
sY <- matrix(Yp, 6, 6) + E * sqrt(matrix(Yp, 6,
6))
if(min(sY[is.na(Y) == FALSE]) >= 0){
sbase <- data.frame(sY = as.vector(sY), lig,
col)
sbase$sY[is.na(Y) == TRUE] = NA
```

Uncertainty in a regression model

```
sreg <- glm(sY ~ lig + col, data = sbase,
family = poisson(link = "log"))
sYp <- predict(sreg, type = "response", newdata
= sbase)
R[s] <- sum(sYp[is.na(Y) == TRUE])
sYpscenario <- rqpoisG(36, sYp, phi = 3.18623)
Rs[s] <- sum(sYpscenario[is.na(Y) == TRUE])
}}
```

Uncertainty in a regression model

- When generating pseudo triangles, negative increments can appear. Indeed, for two values of $\widehat{Y}_{i,j}$, it is possible that $\hat{\varepsilon}\sqrt{\widehat{Y}_{i,j}}$ is negative (if the residue is the smallest obtained residue).

Uncertainty in a regression model

```
sort(residus)[1:2]
       14        13
-3.207593 -2.221573
sort(sqrt(Yp[is.na(Y) == FALSE]))[1:4]
      25       26       19       20
2.868171 2.962026 4.375362 4.518536
```

Uncertainty in a regression model

- The chosen solution is not to take into account the triangles with negative increments, what should we overestimate the lower Quantiles. However, provisioning, lower Quantile s have little interest. Quantile of higher order are estimated here by

Uncertainty in a regression model

```
Rna <- R
Rna[is.na(R) == TRUE] <- 0
Rsna <- Rs
Rsna[is.na(Rs) == TRUE] <- 0
quantile(Rna, c(.75, .95, .99, .995))
     75%      95%      99%    99.5%
2469.491 2602.630 2705.508 2740.274
quantile(Rsna, c(.75, .95, .99, .995))
    75%     95%     99%   99.5%
2495.00 2643.00 2764.01 2809.00
```

Uncertainty in a regression model

- From these 20,000 simulated triangles, we can obtain the distribution of the estimated amounts of provisions (stored in the vector R) but also payments scenarios (and therefore necessary provisions in the Rs vector). 10% of scenarios that resulted in missing values for this one will be deleted.

Uncertainty in a regression model

```
Rnarm <- R[is.na(R) == FALSE]
> Rsnarm <- Rs[is.na(Rs) == FALSE]
> quantile(Rnarm, c(.75, .95, .99, .995))
     75%      95%      99%    99.5%
2477.553 2609.876 2710.583 2743.981
> quantile(Rsnarm, c(.75, .95, .99, .995))
    75%     95%     99%   99.5%
2507.00 2652.00 2772.00 2818.68
```

Uncertainty in a regression model

- Figure 13 and 14 give an estimate of the distribution of $\widehat{R}$ but also R.

```
plot(density(Rnarm), col = "grey", main ="")
lines(density(Rsnarm), lwd = 2)
boxplot(cbind(Rnarm,Rsnarm),col =
c("grey","black"), horizontal = TRUE)
```

Uncertainty in a regression model

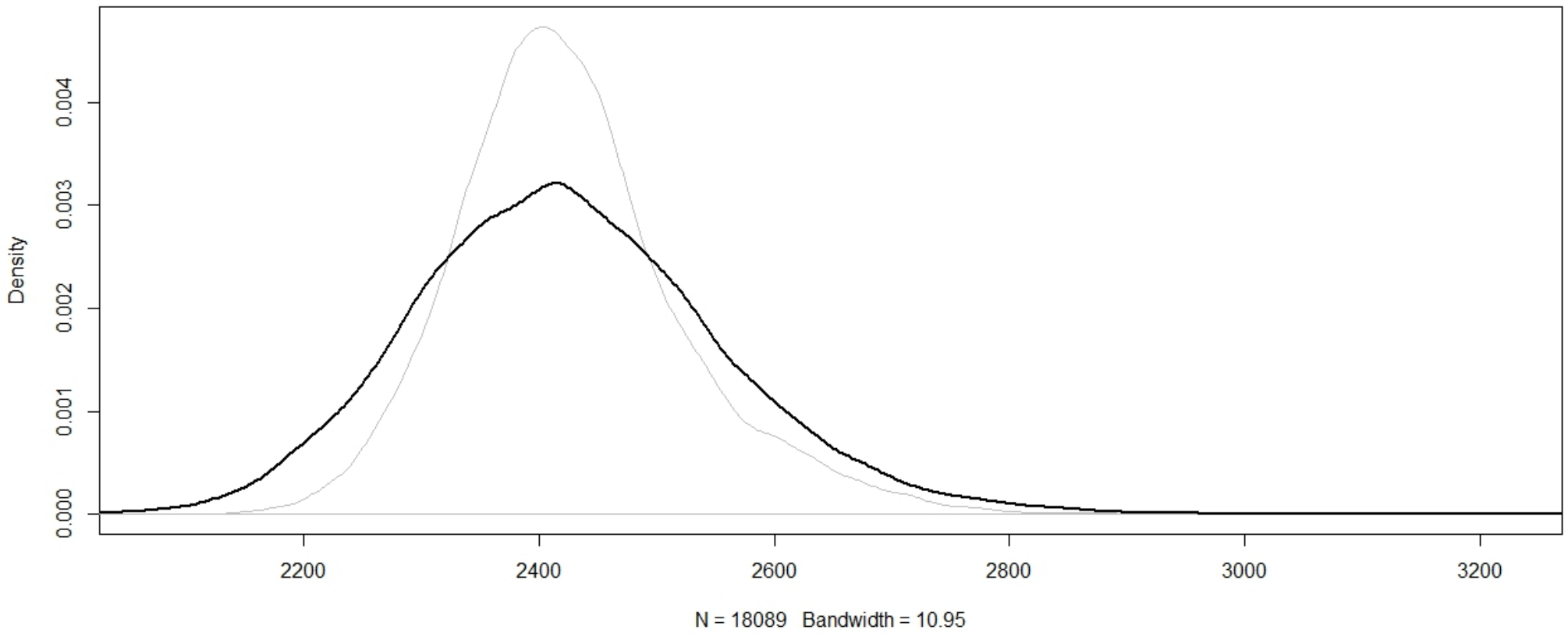


Figure 13 - Estimated density of $\widehat{R}$ and R.

Uncertainty in a regression model

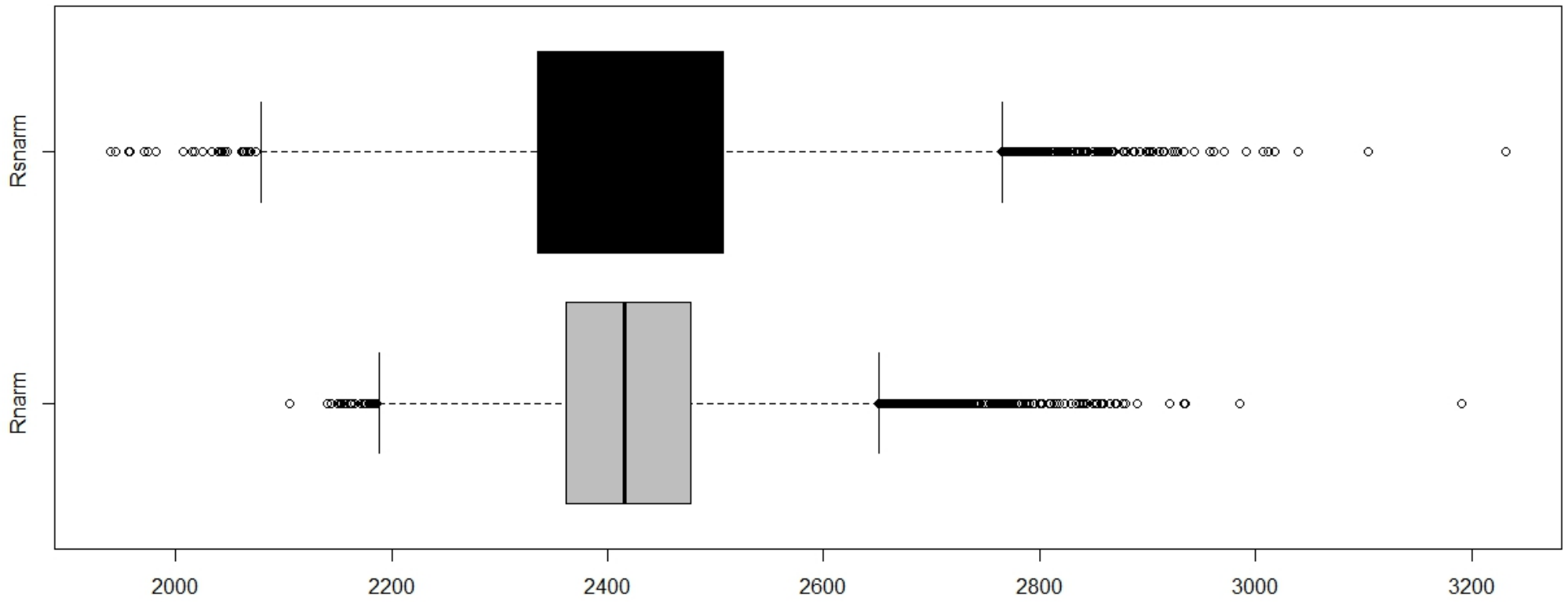


Figure 14 - Boxplot of $\widehat{R}$ and R.

Uncertainty in a regression model

- Note that the quantities obtained are very close to those obtained by the function bootChainLadder of library(ChainLadder).

```
library(ChainLadder)

BootChainLadder(PAID, 20000, "od.pois")
```

[redacted]

	Latest	Mean Ultimate	Mean IBNR	SD IBNR	IBNR 75%	IBNR 95%
1	4,456	4,456	0.0	0.0	0	0
2	4,73	4,752	22.2	12.1	28	45
3	5,42	5,455	35.5	15.2	44	64
4	6,02	6,085	65.4	19.7	77	101
5	6,794	6,946	152.0	28.4	169	201
6	5,217	7,363	2,146.4	111.4	2,217	2,337
	Totals					
Latest:	32,637					
Mean Ultimate:	35,058					
Mean IBNR:	2,421					
SD IBNR:	131					
Total IBNR 75%:	2,503					
Total IBNR 95%:	2,647					

Uncertainty in a regression model

- Two basic parameters are involved in a generalized linear regression.

- The link function, which links to the prediction factors here:

$$\widehat{\boldsymbol{Y}}_{\boldsymbol{i,j}} = \mathbb{E}\left(\boldsymbol{Y}_{\boldsymbol{i,j}} | \boldsymbol{F}_{\boldsymbol{n}}\right) = \boldsymbol{exp}\left[\widehat{\boldsymbol{\gamma}} + \widehat{\boldsymbol{\alpha}}_{\boldsymbol{i}} + \widehat{\boldsymbol{\beta}}_{\boldsymbol{j}}\right]$$

- Law or variance function, which gives the shape of the confidence interval here

$$\mathbb{V}\left(\boldsymbol{Y}_{\boldsymbol{i,j}} | \boldsymbol{F}_{\boldsymbol{n}}\right) = \boldsymbol{\varphi} \times \mathbb{E}\left(\boldsymbol{Y}_{\boldsymbol{i,j}} | \boldsymbol{F}_{\boldsymbol{n}}\right)$$

Uncertainty in a regression model

- The only motivation Poisson model (or quasi-Poisson) is that it provides exactly the same amount as the Chain-Ladder method. But no statistical criteria have been mentioned, for now, to legitimize this model.

Uncertainty in a regression model

- The Tweedie models are an over family model, including the Poissonian model. It is assumed that
- The link function is a power function, or rather a transformed Box-Cox,

$$\widehat{Y}_{i,j} = g_{\lambda}^{-1}\left[\widehat{\gamma} + \widehat{\alpha}_i + \widehat{\beta}_j\right]$$

- Where

$$g_{\lambda}(x) = \lambda^{-1}\left[x^{\lambda} - 1\right]$$

- If λ > 0

Uncertainty in a regression model

- With the limiting case

$$\boldsymbol{g_0(x) = log(x)}$$

- The variance function, which gives the shape of the confidence interval, here

$$\mathbb{V}\left(\boldsymbol{Y_{i,j}|F_n}\right) = \boldsymbol{\varphi} \times \mathbb{E}\left(\boldsymbol{Y_{i,j}|F_n}\right)^{\boldsymbol{\mu}}$$

- Where λ and μ parameters are unknown.

Uncertainty in a regression model

- The density of a law Tweedie of mu parameter is here

```
ftweedie <- function(y, p, mu, phi) {
if(p == 2) {f <- dgamma(y, 1/phi, 1/(phi*mu))}
else
if(p == 1) {f <- dpois(y/phi, mu/phi)} else
{lambda <- mu^(2-p)/phi/(2-p)
if(y == 0) { f <- exp(-lambda)} else
```

Uncertainty in a regression model

```
{ alpha <- (2-p)/(p-1)
beta <- 1 / (phi * (p-1) * mu^(p-1))
k <- max(10, ceiling(lambda + 7 *
sqrt(lambda)))
f <- sum(dpois(1:k, lambda) * dgamma(y, alpha *
(1:k), beta))
}
}
return(f)}
```

Uncertainty in a regression model

- To judge the relevance of the adjustment, we can calculate the log-likelihood of the model, keeping a log link, for example (which is sometimes easier the numeric level, but also in terms of interpretation).

Uncertainty in a regression model

```
pltweedie <- function(puissance) {
regt <- glm(Y ~ lig + col,
tweedie(puissance,0))
reserve <- sum(predict(regt, type =
"response", newdata =
data.frame(lig,col))[is.na(Y) == TRUE])
dev <- deviance(regt)
phi <- dev/n
```

Uncertainty in a regression model

```
mu <- predict(regt, type = "response",
newdata = data.frame(lig,col))
hat.logL <- 0
for (k in which(is.na(Y) == FALSE)) {
hat.logL <- hat.logL + log(ftweedie(Y[k],
puissance, mu[k], phi)) }
return(list(puissance = puissance, phi = phi,
reserve = reserve, logL = hat.logL))
}
```

Uncertainty in a regression model

- If we calculate the log-likelihood values for 5, between 1 and 2 (corresponding to the case of a Poisson regression and Gamma regression) we obtain

```
library(statmod)
```

Uncertainty in a regression model

```
pltweedie(puissance=1.25)
$puissance
[1] 1.25
$phi
[1] 0.4660584
$reserve
[1] 2426.639
$logL
[1] -95.96533
```

Uncertainty in a regression model

```
pltweedie(puissance=1.5)
$puissance
[1] 1.5
$phi
[1] 0.1547666
$reserve
[1] 2428.269
$logL
[1] -99.20657
```

Uncertainty in a regression model

- Figure 15 shows the influence of the power parameter of the variance function. Clearly, the likelihood is maximal for a power close to 1 (which corresponds to the Poisson ele model) with a logarithmic link,

```
puiss <- seq(1.02,1.98,by=.01)

TW <- function(p){pltweedie(p)$logL}
```

Uncertainty in a regression model

```
optimize(TW, c(1.01,1.99), tol=1e-4,maximum = TRUE)
$maximum
[1] 1.01005
$objective
[1] -92.1735
plot(puiss,Vectorize(TW)(puiss),type="l", xlab="Puissance de la loi Tweedie",ylab="log vraisemblance")
```

Uncertainty in a regression model

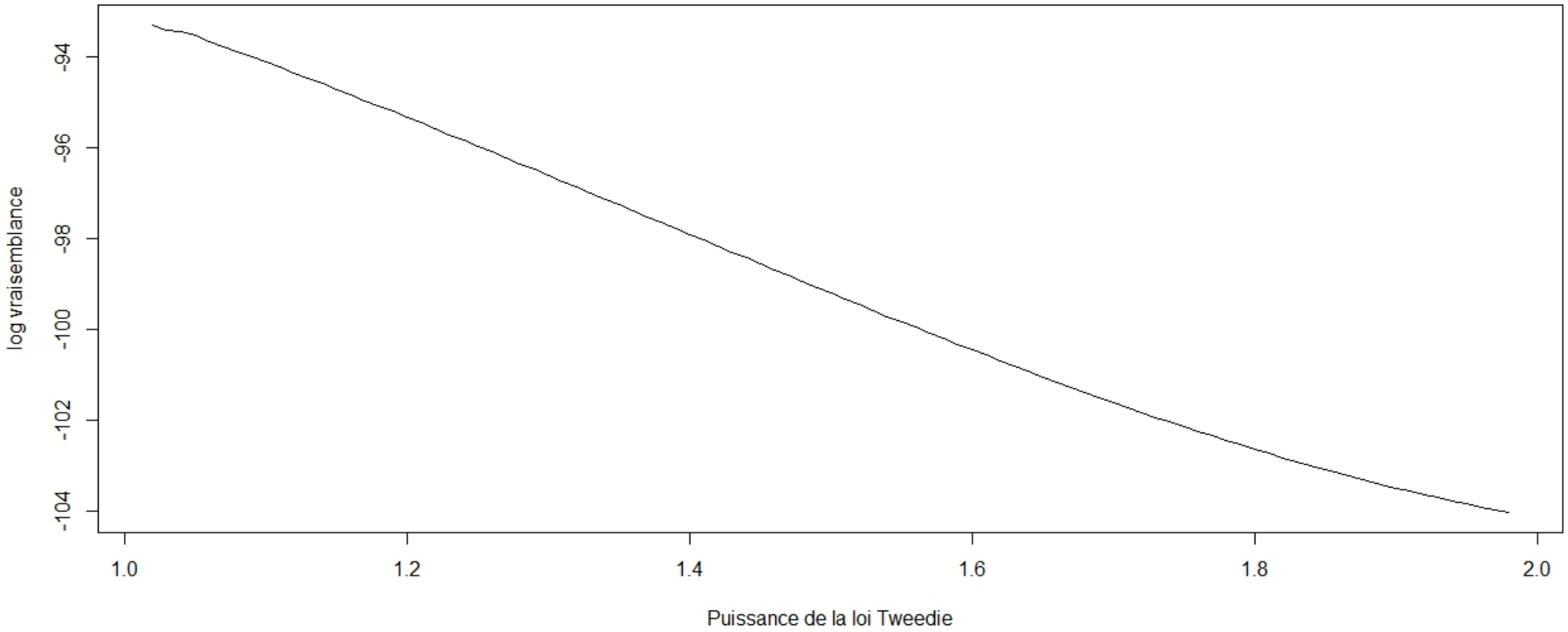


Figure 15 - Evolution of the log-likelihood profile as a function of μ.

Multivariate triangles

- As was explained in the introduction, the use of triangles, and methods of payment rates, is only possible if the triangles are stable and homogeneous.

Multivariate triangles

- But it is not uncommon that a triangle has relatively different risks in their development. For example, car insurance, property and bodily injury are significantly different.

Multivariate triangles

- Inspired by the idea of Mack, we can assume that

$$\hat{R}_i \sim LN(\mu_i, \sigma_i^2)$$

- For i = 1,2.

- If the independent risk is assumed, the law of the sum is simply the convolution of the two laws. We can use distribution to families S4 format and library(distr).

Multivariate triangles

- Recall that if

$$\boldsymbol{X \sim LN(\mu, \sigma^2)}$$

$$\boldsymbol{\mu = log[\mathbb{E}(X)] - \frac{1}{2} log\left(1 + \frac{\mathbb{V}(X)}{\mathbb{E}(X)^2}\right)}$$

$$\boldsymbol{\sigma^2 = log\left(1 + \frac{\mathbb{V}(X)}{\mathbb{E}(X)^2}\right)}$$

Multivariate triangles

- From the means and variances, we deduce the laws of both amounts of provisions. Assuming that the two triangles are independent, then

```
P.corp <-
matrix(c(3209,4372,4411,4428,4435,4456,3367,465
9,4696,4720,4730,NA,3871,5345,5338,5420,NA,NA,4
239,5917,6020,NA,NA,NA,4929,6794,NA,NA,NA,NA,52
17,NA,NA,NA,NA,NA), nrow = 6, ncol = 6, byrow =
T)
```

Multivariate triangles

```
P.mat <-
matrix(c(3340,4590,4450,4550,4750,4600,3500,450
0,4700,4990,4950,NA,4000,5400,5500,5700,NA,NA,4
600,6200,6500,NA,NA,NA,5300,7000,NA,NA,NA,NA,55
00,NA,NA,NA,NA,NA), nrow = 6, ncol = 6, byrow =
T)
```

Multivariate triangles

```
library(distr)
n = nrow(P.mat)
library(ChainLadder)
V = MackChainLadder(P.mat)$Total.Mack.S.E^2
E = sum(MackChainLadder(P.mat)$FullTriangle[,n]
-
diag(MackChainLadder(P.mat)$FullTriangle[n:1,])
)
mu = log(E) - .5*log(1+V/E^2)
sigma2 = log(1+V/E^2)
```

Multivariate triangles

```
LM = Lnorm(meanlog = mu, sdlog = sqrt(sigma2))
V = MackChainLadder(P.corp)$Total.Mack.S.E^2
E =
sum(MackChainLadder(P.corp)$FullTriangle[,n] -
diag(MackChainLadder(P.corp)$FullTriangle[n:1,]
))
mu = log(E) - .5*log(1+V/E^2)
sigma2 = log(1+V/E^2)
LC = Lnorm(meanlog = mu, sdlog = sqrt(sigma2))
LT = LM + LC
```

Multivariate triangles

- We can then compare the convolution law and lognormal cumulative adjusted on the triangle,

```
P.tot = P.mat + P.corp
V = MackChainLadder(P.tot)$Total.Mack.S.E^2
E  =  sum(MackChainLadder(P.tot)$FullTriangle[,n]
-
diag(MackChainLadder(P.tot)$FullTriangle[n:1,])
)
mu = log(E) - .5*log(1+V/E^2)
sigma2 = log(1+V/E^2)
```

Multivariate triangles

- Figure 16 compares the distribution obtained by convolant two lognormal laws (assuming independence between the triangles, and the amount of allowance can be modeled by a lognormal distribution) and the distribution of the amount of provision obtained by aggregating two payments of triangles.

Multivariate triangles

```
u = seq(E - 4*sqrt(V), E + 4*sqrt(V), length =
101)
vtotal = dlnorm(u,mu,sqrt(sigma2))
vconvol = d(LT)(u)
plot(u,vconvol,col="grey",type="l",xlab="montan
t de provision",ylab="")
lines(u,vtotal,lwd=2)
legend(470000,1.2e-05,c("convolution","somme
des
triangles"),col=c("grey","black"),lwd=c(1,2),bt
y="n")
```

Multivariate triangles

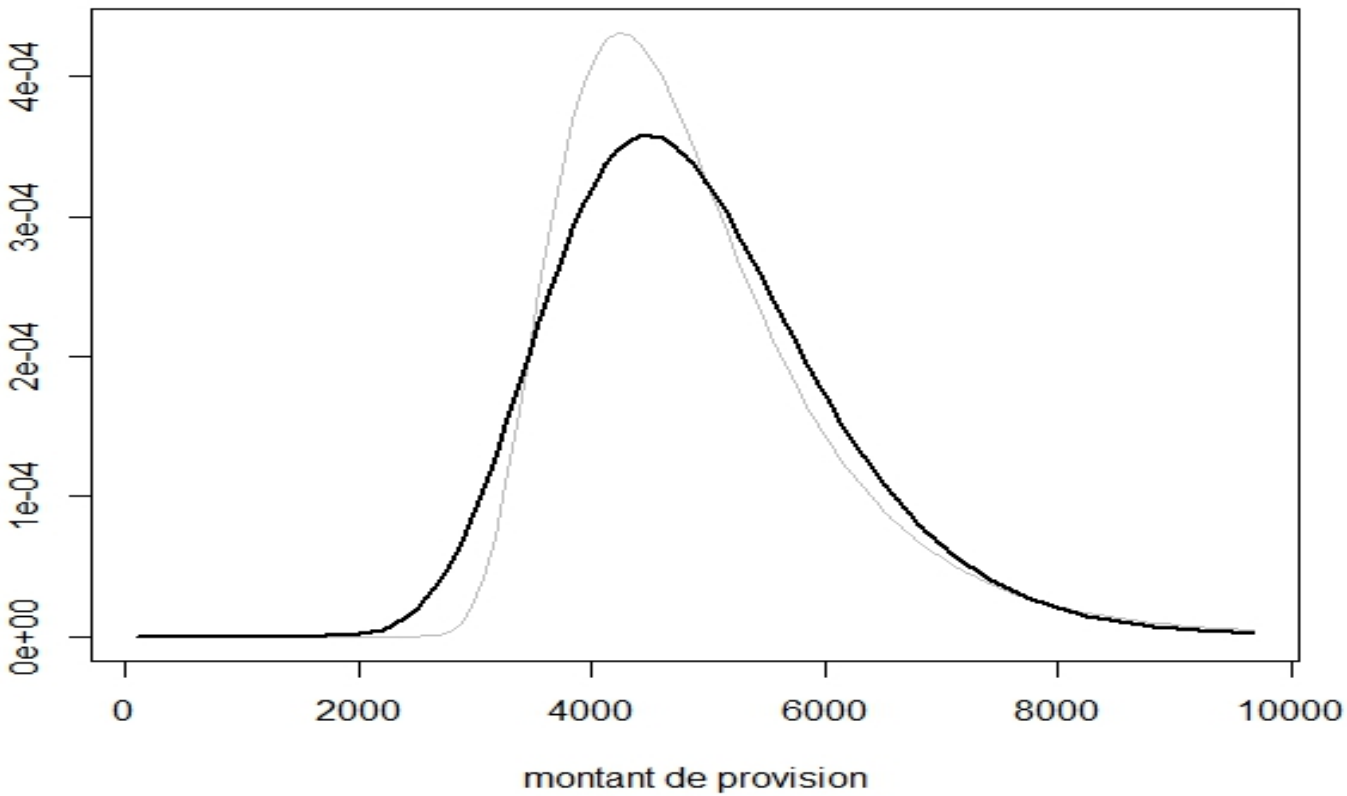


Figure 16 - Distribution of the total amount of provisions, by summing the provisions per triangles and working on the aggregated triangle.

Multivariate triangles

- 95% quantile are then respectively.

```
q(LT)(.95)

[1] 7166.105

qlnorm(.95,mu,sqrt(sigma2))

[1] 7066.893
```

Multivariate triangles

- For the convolution law and the sum of the two triangles. Two interpretations are possible: triangles supposed to be independent is probably too strong hypothesis and work on an aggregated triangle (and therefore not homogeneous) introduces an additional uncertainty.

Multivariate triangles

- Pröhl & Schmidt proposed a method of Chain-Ladder in a multivariate framework. We note

$$\boldsymbol{\lambda_{i,j}} = \left(\boldsymbol{\lambda_{i,j}^{(k)}}\right)$$

- Where

$$\boldsymbol{\lambda_{i,j}^{(k)}} = \frac{\boldsymbol{C_{i,j}^{(k)}}}{\boldsymbol{C_{i,j-1}^{(k)}}}$$

Multivariate triangles

- And

$$\boldsymbol{C_{i,j}} = \left(\boldsymbol{C_{i,j}^{(k)}}\right) \in \mathbb{R}^{\boldsymbol{k}}$$

- It is assumed that there is

$$\boldsymbol{\lambda_j} \in \mathbb{R}^{\boldsymbol{k}}$$

$$\mathbb{E}\left[\boldsymbol{C_{i,j}} | \boldsymbol{C_{i,j-1}}\right] = \boldsymbol{diag}\left(\boldsymbol{\lambda_{j-1}}\right) \times \boldsymbol{C_{i,j-1}}$$

Multivariate triangles

- And

$$\boldsymbol{Cov}\left[\boldsymbol{C_{i,j}}, \boldsymbol{C_{i,j}} | \boldsymbol{C_{i,j-1}}\right] = \boldsymbol{diag}\left(\sqrt{\boldsymbol{C_{i,j-1}}}\right) \times \sum_{\boldsymbol{j-1}} \times \boldsymbol{diag}\left(\sqrt{\boldsymbol{C_{i,j-1}}}\right)$$

- So in these assumptions as in the univariate case, we can write

$$\mathbb{E}\left[\boldsymbol{C_{i,n}} | \boldsymbol{C_{i,n-i}}\right] = \prod_{\boldsymbol{j=n-i}}^{\boldsymbol{n-1}} \boldsymbol{diag}(\boldsymbol{\lambda_j}) \times \boldsymbol{C_{i,n-i}}$$

Multivariate triangles

- The estimator of the transition factor is

$$\hat{\boldsymbol{\lambda}}_{\boldsymbol{j}} = \left[\sum_{\boldsymbol{i=0}}^{\boldsymbol{n-j-1}} \boldsymbol{diag}\left(\sqrt{\boldsymbol{C}_{\boldsymbol{i,j-1}}}\right) \times \sum_{\boldsymbol{j}}^{-\boldsymbol{1}} \times \boldsymbol{diag}\left(\sqrt{\boldsymbol{C}_{\boldsymbol{i,j-1}}}\right)\right]^{-\boldsymbol{1}} \times \sum_{\boldsymbol{i=0}}^{\boldsymbol{n-j-1}} \boldsymbol{diag}\left(\sqrt{\boldsymbol{C}_{\boldsymbol{i,j-1}}}\right) \times \sum_{\boldsymbol{j}}^{-\boldsymbol{1}} \times \boldsymbol{diag}\left(\sqrt{\boldsymbol{C}_{\boldsymbol{i,j-1}}}\right) \times \boldsymbol{\lambda}_{\boldsymbol{i,j+1}}$$

Multivariate triangles

- The Chain-Ladder estimator is the ultimate charge

$$\widehat{\boldsymbol{C}}_{\boldsymbol{i},\boldsymbol{n}} = \prod_{\boldsymbol{j}=\boldsymbol{n}-\boldsymbol{i}}^{\boldsymbol{n}-\boldsymbol{1}} \boldsymbol{diag}\left(\widehat{\boldsymbol{\lambda}}_{\boldsymbol{j}}\right) \times \boldsymbol{C}_{\boldsymbol{i},\boldsymbol{n}-\boldsymbol{i}}$$

- This estimator checks the same properties as in the univariate case. In particular, this estimator is an unbiased estimator of [Ci,n|Ci,n-i] but also $\mathbb{E}$[Ci,n].

- It is also possible to calculate the prediction mse.

Multivariate triangles

- The Chain-Ladder estimator is the ultimate charge

$$\widehat{\boldsymbol{C}}_{\boldsymbol{i},\boldsymbol{n}} = \prod_{\boldsymbol{j}=\boldsymbol{n}-\boldsymbol{i}}^{\boldsymbol{n}-\boldsymbol{1}} \boldsymbol{diag}\left(\widehat{\boldsymbol{\lambda}}_{\boldsymbol{j}}\right) \times \boldsymbol{C}_{\boldsymbol{i},\boldsymbol{n}-\boldsymbol{i}}$$

- This estimator checks the same properties as in the univariate case. In particular, this estimator is an unbiased estimator of [Ci,n|Ci,n-i] but also $\mathbb{E}$[Ci,n].

- It is also possible to calculate the prediction mse.

Multivariate triangles

- The idea in econometric models is to assume that residues can be correlated,

```
ligne = rep(1:n, each = n); colonne = rep(1:n,
n)
PAID = P.corp; INC = PAID
INC[,2:n] = abs(PAID[,2:n] - PAID[,1:(n-1)])
I.corp = INC
PAID = P.mat; INC = PAID
INC[,2:n] = abs(PAID[,2:n] - PAID[,1:(n-1)])
```

Multivariate triangles

```
I.mat = INC
Ym = as.vector(I.mat)
Yc = as.vector(I.corp)
lig = as.factor(ligne)
col = as.factor(colonne)
base = data.frame(Ym,Yc,col,lig)
```

Multivariate triangles

```
regm = glm(Ym ~ col + lig, data = base, family
= "poisson")
regc = glm(Yc ~ col + lig, data = base, family
= "poisson")
res.corp = residuals(regc, type = "pearson")
res.mat = residuals(regm, type = "pearson")
cor(res.corp,res.mat)
[1] 0.274203
```

- Note as well that the correlation is not null.

Multivariate triangles

- Once noted that there are probably a dependency between the two triangles, it seems legitimate to take into account in the simulation algorithms discussed in the previous section.

Multivariate triangles

- For the estimation error, when pulling the residues, does not take them independently in the two triangles. One then pulls the pairs of residues

$$\hat{\boldsymbol{\varepsilon}}_{\boldsymbol{i,j}}^{\boldsymbol{matériel,b}}, \hat{\boldsymbol{\varepsilon}}_{\boldsymbol{i,j}}^{\boldsymbol{corporel,b}}$$

- For error, one can draw a bivariate Poisson if using a bivariate regression (implemented in library(bivpois) or vector Gaussian bivariate).

Multivariate triangles

- In the second case

$$\begin{pmatrix} C_{i,j+1}^{matériel} \\ C_{i,j+1}^{corporel} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \lambda_j^m C_{i,j}^{matériel} \\ \lambda_j^c C_{i,j}^{corporel} \end{pmatrix}, \begin{pmatrix} \sigma_j^{m2} C_{i,j}^{matériel} & * \\ * & \sigma_j^{c2} C_{i,j}^{corporel} \end{pmatrix} \right)$$

Borhutter-Fergusson Benktander and Bayesian Methods

- The first two methods that we will see often been proposed as an alternative to the chain ladder method because they introduce a priori on the ultimate load.

Borhutter-Fergusson Benktander and Bayesian Methods

- Traditionally, here we continue to assume that
  - Accident years are independent of each other
  - There are μi and development factors β1, β2, …, βn - with βn = 1 - such as

$$\mathbb{E}\left(C_{i,1}\right) = \beta_1 \mu_i$$

$$\mathbb{E}\left(C_{i,j+k} | C_{i,1}, \dots, C_{i,j}\right) = C_{i,j} + \left[\beta_{j+k} - B_j\right]\mu_i$$

Borhutter-Fergusson Benktander and Bayesian Methods

- Under these assumptions, for all i, j,

$$\mathbb{E}\left(\boldsymbol{C_{i,j}}\right) = \boldsymbol{\beta_j \mu_i}$$

- That reminds factor models mentioned earlier. Except here only

$$\boldsymbol{\beta} = (\boldsymbol{\beta_1}, \boldsymbol{\beta_2}, \dots, \boldsymbol{\beta_n})$$

Borhutter-Fergusson Benktander and Bayesian Methods

- Will estimate statistically,

$$\boldsymbol{\mu} = \widehat{\boldsymbol{\mu}}_{\boldsymbol{i}}$$

- Being obtained by expert opinion, $\widehat{\boldsymbol{\mu}}_{\boldsymbol{i}}$ is an estimator of

$$\mathbb{E}\left(\boldsymbol{C}_{\boldsymbol{i},\boldsymbol{n}}\right)$$

Borhutter-Fergusson Benktander and Bayesian Methods

- With these two estimates, we deduce the estimator of

$$\mathbb{E}\left(C_{i,j+k} | C_{i,1}, \dots, C_{i,j}\right)$$

- Of the form

$$\widehat{C}_{i,n} = C_{i,j} + \left[1 - \widehat{\beta}_{j-i}\right]\widehat{\mu}_i$$

Borhutter-Fergusson Benktander and Bayesian Methods

- The estimator proposed by Bornhutter-Ferguson is then simply derived from the Chain-Ladder method, laying

$$\widehat{\boldsymbol{\beta}}_j = \prod_{k=j+1}^{n} \frac{1}{\widehat{\lambda}_k}$$

Borhutter-Fergusson Benktander and Bayesian Methods

- Finally, to estimate μ is assumed to have a loss ratio (sinister/premium) target, for example 105%, occurrence year. Under these conditions, we can then simply estimate the amount of provisions,

Borhutter-Fergusson Benktander and Bayesian Methods

```
PREMIUM <- c(4591, 4672, 4863, 5175, 5673,
6431)
mu <- 1.05 * PREMIUM
beta <- rev(cumprod(rev(1/LAMBDA)))
Cdiag <- diag(PAID[,nc:1])
Cultime <- Cdiag + (1 - c(1, rev(beta))) * mu
Cultime - Cdiag
sum(Cultime - Cdiag)
```

Borhutter-Fergusson Benktander and Bayesian Methods

- The estimator of Benktander, taken a few years later by Hovinen, based on an a priori estimator of the ultimate load Ci,n, noted µi. It is also assumed that there is a payment rate

$$\boldsymbol{\beta} = (\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \dots, \boldsymbol{\beta}_n)$$

Borhutter-Fergusson Benktander and Bayesian Methods

- Known, such as

$$\mathbb{E}\left(\boldsymbol{C_{i,j}}\right) = \boldsymbol{\beta_j \mu_i}$$

- Under these assumptions, the amount of provision should be

$$\widehat{\boldsymbol{R}}_{\boldsymbol{i}} = \widehat{\boldsymbol{C}}_{\boldsymbol{i,n}} - \boldsymbol{C_{i,n-i}} = [\mathbf{1} - \boldsymbol{\beta_{n-i}}]\boldsymbol{\mu_i}$$

Borhutter-Fergusson Benktander and Bayesian Methods

- It is using the Chain-Ladder estimator, averaged with the a priori estimate of the ultimate load. then

$$\widehat{R}_i^{BH} = \widehat{C}_{i,n} - C_{i,n-i} = [1 - \beta_{n-i}]\left(\beta_{n-i}\widehat{C}_{i,n}^{CL} + [1 - \beta_{n-i}]\mu_i\right)$$

Borhutter-Fergusson Benktander and Bayesian Methods

- Note that

$$\widehat{R}_i^{BH} = (1 - \beta_{n-i})\widehat{C}_i^{BF}$$

- If the cadence

$$\beta = (\beta_1, \beta_2, \dots, \beta_n)$$

Borhutter-Fergusson Benktander and Bayesian Methods

- Is constructed from the development factors induced by Chain-Ladder method. Another writing of this term is to write the ultimate load (and not the amount of provision).

$$\widehat{C}_i^{BH} = C_{i,n-i} + (1 - \beta_{n-i})\widehat{C}_i^{BF} = \beta_{n-i}\widehat{C}_i^{CL} + (1 - \beta_{n-i})\widehat{C}_i^{BF}$$

- This allows seeing the prediction of Benktander as a convex combination of Chain-Ladder estimators and Bornhuetter-Ferguson.

Borhutter-Fergusson Benktander and Bayesian Methods

- In this approach, there is used another expert opinion. The idea is to rewrite the expression

$$C_{i,n} = C_{i,n-i} + \left(1 - \frac{C_{i,n-i}}{C_{i,n}}\right) C_{i,n}$$

Borhutter-Fergusson Benktander and Bayesian Methods

- In the form

$$C_{i,n} = C_{i,n-i} + \left(1 - \frac{C_{i,n-i}}{C_{i,n}}\right) LR_i \times P_i$$

Borhutter-Fergusson Benktander and Bayesian Methods

- Where LRi corresponds to loss ratio for the year i, i.e.

$$LR_i = \frac{C_{i,n}}{P_i}$$

Borhutter-Fergusson Benktander and Bayesian Methods

- The idea of the Cape Code-known method is to write a more general form,

$$\boldsymbol{C_{i,n} = C_{i,n-i} + (1 - \pi_{n-i})LR_i \times P_i}$$

- Where π_{n-i} corresponds to a rate of payment, and can be estimated by the Chain-Ladder method. As for LRi it comes to loss ratio targets, corresponding to an expert opinion.

Borhutter-Fergusson Benktander and Bayesian Methods

- We can also offer the same target ratio for several occurrence years. We will then ask

$$\boldsymbol{R_i = C_{i,n} - C_{i,n-i} = (1 - \pi_{n-i}) LR_{\mathcal{A}} \times P_i}$$

- For $i \in \mathcal{A}$, where

$$\boldsymbol{LR_{\mathcal{A}} = \frac{\sum_{k \in \mathcal{A}} C_{n,n-k}}{\sum_{k \in \mathcal{A}} \pi_{n-k} P_k}}$$

Borhutter-Fergusson Benktander and Bayesian Methods

- First, one can calculate πi from the Chain-Ladder method, i.e.

$$\boldsymbol{\pi_{n-i} = \frac{C_{i,n-i}}{C_{i,n}}}$$

- Where the ultimate burden that is predicted by the Chain-Ladder method.

Borhutter-Fergusson Benktander and Bayesian Methods

```
Cultime =
MackChainLadder(PAID)$FullTriangle[,nc]
PREMIUM <- c(4591, 4672, 4863, 5175, 5673,
6431)
mu <- 1.05 * PREMIUM
beta <- rev(cumprod(rev(1/LAMBDA)))
Cdiag <- diag(PAID[,nc:1])
Cultime <- Cdiag + (1 - c(1, rev(beta))) * mu
Cultime - Cdiag
```

Borhutter-Fergusson Benktander and Bayesian Methods

```
Cultime - Cdiag
sum(Cultime - Cdiag)
library(ChainLadder)
Cultime =
MackChainLadder(PAID)$FullTriangle[,nc]
(PI <- (1 - Cdiag / Cultime))
```

Borhutter-Fergusson Benktander and Bayesian Methods

```
LR <- TRIANGLE[,nc] / PREMIUM
Cdiag <- diag(PAID[,nc:1])
(Cultime - Cdiag) / (LR * PREMIUM)
```

Borhutter-Fergusson Benktander and Bayesian Methods

- If we then assume that

$$\mathcal{A} = \{1, 2, \dots, n\}$$

- Then

Borhutter-Fergusson Benktander and Bayesian Methods

```
LR = sum(TRIANGLE[,6]) / sum(PREMIUM)
PI * LR * PREMIUM
sum(PI * LR * PREMIUM)
```

- We obtained by a lower amount of total provision to that obtained by the chain ladder method, since the amount of provisions here is 2,357.756.

Borhutter-Fergusson Benktander and Bayesian Methods

- Bayesian approaches have been popularized in Actuarial Science by the method of credibility, corresponding to a Bayesian approach in a linear framework. But it is possible to go further.

Borhutter-Fergusson Benktander and Bayesian Methods

- Classically, suppose we are interested to Y whose law would be f(.|θ), where very generally,

$$\boldsymbol{Y} = \left(\boldsymbol{Y}_{\boldsymbol{i},\boldsymbol{j}}\right)$$

- And

$$\boldsymbol{\theta} = \left(\boldsymbol{\theta}_{\boldsymbol{i},\boldsymbol{j}}\right)$$

Borhutter-Fergusson Benktander and Bayesian Methods

- Y can be here the combined payments of C triangle, the triangle Y increments, or triangle transition ratios payments rates

$$\lambda = \frac{C_{i,j+1}}{C_{i,j}}$$

Borhutter-Fergusson Benktander and Bayesian Methods

- In Mack's approach attempts to model Y can be here the triangle of cumulative payments C, and

$$\boldsymbol{\theta_j} = \left(\boldsymbol{\lambda_j}, \boldsymbol{\sigma_j^2}\right)$$

Borhutter-Fergusson Benktander and Bayesian Methods

- Here we focuses on the λ law, which depend on θ = (θj), where θj = (γj,σj2), where, for simplicities ratings (and avoid mixing with λi,j) there is γj the underlying factor of development.

$$\lambda_{i,j} | \left(\gamma_j, \sigma_j^2\right) \sim \mathcal{N}\left(\gamma_j, \frac{\sigma_j^2}{C_{i,j}}\right)$$

Borhutter-Fergusson Benktander and Bayesian Methods

- Here, σ2 are the parameters of interest, and are supposed to estimated separately (as we already do in Generalized Linear Models). As for Ci,j, they are interpreted here as weights, and are assumed to be known. The log-likelihood is here

$$\boldsymbol{log\mathcal{L}(\lambda|\gamma) = \sum_{i,j} \frac{1}{2}\left(log\left[\frac{C_{i,j-1}}{\sigma_j^2}\right] - \frac{C_{i,j-1}}{\sigma_j^2}\left[\lambda_{i,j} - \gamma_j\right]^2 \right)}$$

Borhutter-Fergusson Benktander and Bayesian Methods

- Using Bayes' formula, the conditional log-γ for λ is simply the density

$$\boldsymbol{log}[\boldsymbol{g}(\boldsymbol{\lambda}|\boldsymbol{\gamma})] = \boldsymbol{log}[\boldsymbol{\pi}(\boldsymbol{\gamma})] + \boldsymbol{log}[\boldsymbol{\mathcal{L}}(\boldsymbol{\lambda}|\boldsymbol{\gamma})] + \boldsymbol{constante}$$

- Where π(.) Is a prior distribution of γ (eg a Gaussian vector).